ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

AERODYNAMIC ANALYSIS OF A VTOL RC AIRCRAFT LIFT FORCE DURING THE TRANSITION FROM VERTICAL TO HORIZONTAL FLIGHT

NUMERICAL FLOW SIMULATION ME-474

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CHAPTER 1

INTRODUCTION

1.1 CONTEXT AND GOALS OF THE STUDY

The objective of this project is to analyze the lift generated by a NACA 4418 wing, with a chord length of $250\ \mathrm{mm}$ and a wingspan of $1500\ \mathrm{mm}$, on a drone VTOL (Vertical Take-Off and Landing) RC aircraft during the transition from vertical to horizontal flight. The VTOL is also equipped with four propellers type tri-blade T5147 which provide high maneuverability in confined and restricted environments. In addition, a front-mounted propeller is used during airplane-mode flight to ensure efficient propulsion and extended operational range over long distances.

A distinctive feature of this VTOL configuration is the presence of retractable wings, shown if Figure 1.1. This design enables the aircraft to combine the advantages of a conventional multirotor drone (namely agility and compactness during vertical flight) with those of a fixed-wing aircraft, offering improved aerodynamic efficiency and a significantly larger operational radius during horizontal flight.

(A) VTOL with folded wings

(B) VTOL with unfolded wings

FIGURE 1.1
The VTOL RC aircraft.

The wing simulations are performed in a two-dimensional plane passing through the axes of rotation of the propellers, which are located on the same side of the aircraft. The purpose of this project is to determine the throttle percentage to be applied to the propellers in order to maintain, at all times during the flight transition, a lift force equal to the weight of the drone. The simulations focus on the analysis of the impact of the propellers on the lift generated by the portion of the wing located directly downstream of the front propeller, for different horizontal flight velocities.

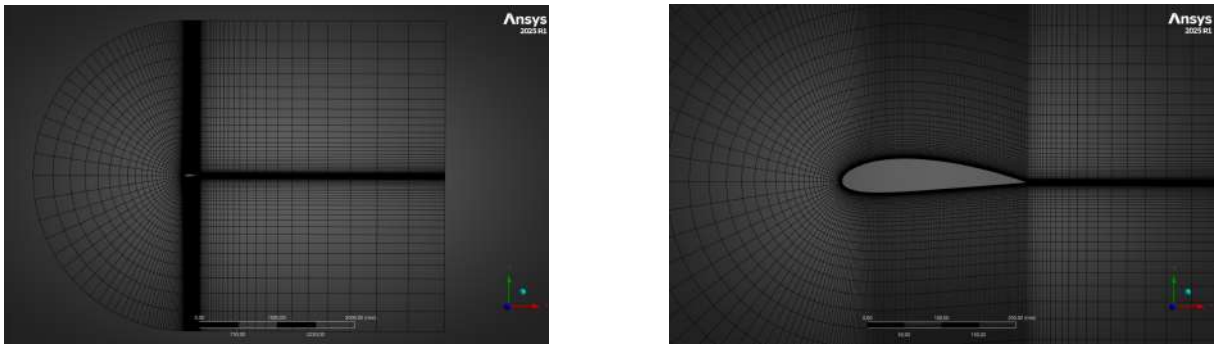
In other words, the objective is to try to satisfy the following force balance :

$$L_1 + L_2 + T = Mg. \tag{1}$$

where L_1 denotes the lift generated by the portion of the wing not affected by the propellers, L_2 denotes the lift generated by the portion of the wing disturbed by the propellers, T represents the thrust produced by the propellers and M is the weight of the drone, and it is estimated to be $2\ \mathrm{kg}$. Therefore, if this equation is satisfied, a stable flight transition can be ensured by adjusting the propeller control based on the results obtained from the simulations. A clear and well-documented schematic to visualize L_1, L_2 and T will be presented in Section 2.5

EXPLANATION ABOUT THE LIFT L_1

The lift contribution L_1 was computed using a mesh similar to the one employed in the present study, as illustrated in Figure 1.2, with the only difference being the absence of the two lines corresponding to the propeller boundary conditions. This mesh has 46 749 nodes and, 46 125 elements. All the modeling choices and numerical settings described in Chapters 2 to 6 were applied in the same, or nearly the same, manner for the computation of L_1.



(A) (B)

FIGURE 1.2
The mesh used to compute the L_1 lift.

This mesh and numerical setup were validated by reproducing the experimental lift coefficient curve reported in the *NACA Report No. 824, Summary of Airfoil Data* (Abbott, Doenhoff and Stivers 1945), using the NACA 4418 airfoil, which is the same airfoil considered in the present project. The relative errors were computed for four different angles of attack (AOA), and the corresponding results are presented below.

TABLE 1.1
Relative error of 2D lift coefficient

AOA	Experimental	Simulation	Relative error (%)
2	0.615459	0.639200	+3.86
4	0.806835	0.834500	+3.43
6	1.002766	1.050000	+4.71
8	1.175915	1.226500	+4.30

Mean absolute relative error on C_l: 4.07 %

1.2 TYPE OF ANALYSIS AND METHODOLOGY

The methodology adopted for the analysis of the vertical-to-horizontal flight transition is as follows. The upstream velocity range is discretized into four representative values, namely $U_\infty = \{3, 6, 9, 12\}\ \mathrm{m/s}$. For each of these velocities, steady-state simulations are performed in order to determine the required control input, *i.e.*, the propeller throttle percentage, such that Equation (1) is satisfied.

As an illustrative example, for an upstream velocity of $U_\infty = 3\ \mathrm{m/s}$, the procedure consists in initially prescribing a trial throttle value for the propellers and evaluating the resulting aerodynamic forces. If the chosen throttle setting does not satisfy Equation (1), that is, if the combined contribution of the lift L_2 and the thrust T, as computed through the simulations, is insufficient, a higher throttle percentage is tested. This iterative process is repeated until the throttle value that maintains the lift as close as possible to the required value is identified or in case an equilibrium condition cannot be met we continue until we reached a satisfying understanding of the dynamics of the model. The same procedure is applied independently for each upstream velocity of 3, 6, 9, and $12\ \mathrm{m/s}$ and all simulations are performed using the finest mesh in order to minimize discretization errors.

Finally, a mesh convergence study is performed for the cases with upstream velocities of $9\ \mathrm{m/s}$ and $12\ \mathrm{m/s}$ in order to select the optimal mesh among the seven meshes generated. The selected mesh corresponds to the one for which the relative error with respect to the finest mesh is sufficiently small (typically below 1%) while keeping the total number of elements as low as possible.

The propellers were modeled using the standard Fluent fan boundary condition (explained in details in Section 4.1.4) in polynomial mode, and the required equation was derived from a formulation found in the literature.

CHAPTER 2

GEOMETRIC MODELING AND HYPOTHESES

2.1 THE GEOMETRY

As shown in Figure 2.1, the computational domain is sized sufficiently large to allow the flow to develop as much as possible and to capture the relevant flow physics as accurately as possible. Upstream of the airfoil, the domain extends over a distance equal to ten times the airfoil chord, while downstream it extends over approximately twenty times the chord.

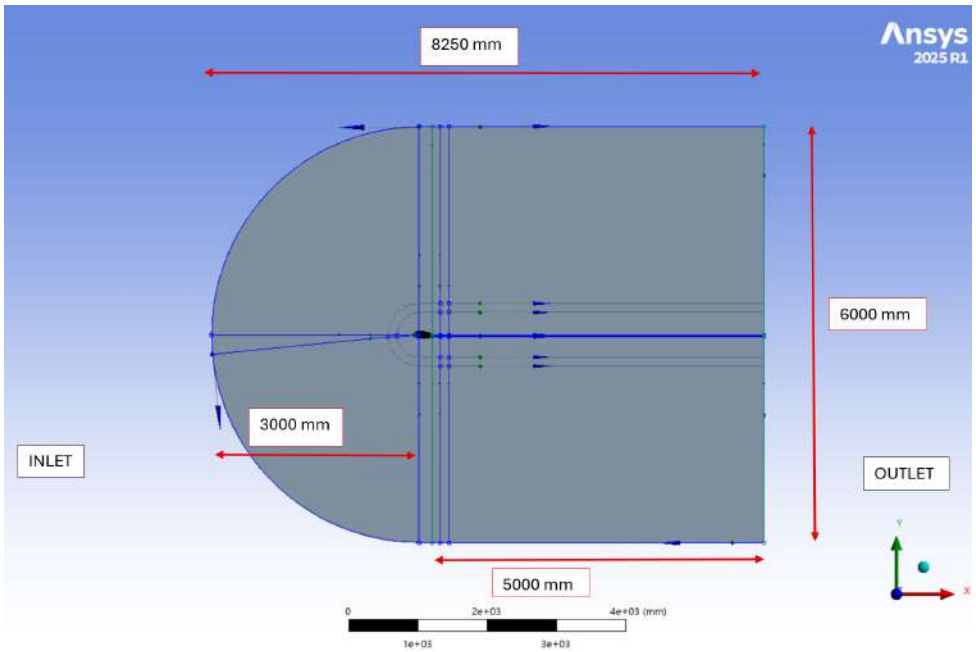


FIGURE 2.1
The geometry of the plane passing through the axis of rotation of the propellers.

The airfoil geometry in Figure 2.2 corresponds to a NACA 4418 airfoil with a chord length of 250 mm and has an angle of attack of 4 degrees. The trailing edge of the airfoil was truncated in order to avoid an excessively sharp geometry, which would otherwise lead to very small cells and poor mesh quality in this

region. This modification improves numerical stability and mesh robustness while having a negligible impact on the global aerodynamic behavior of the flow.

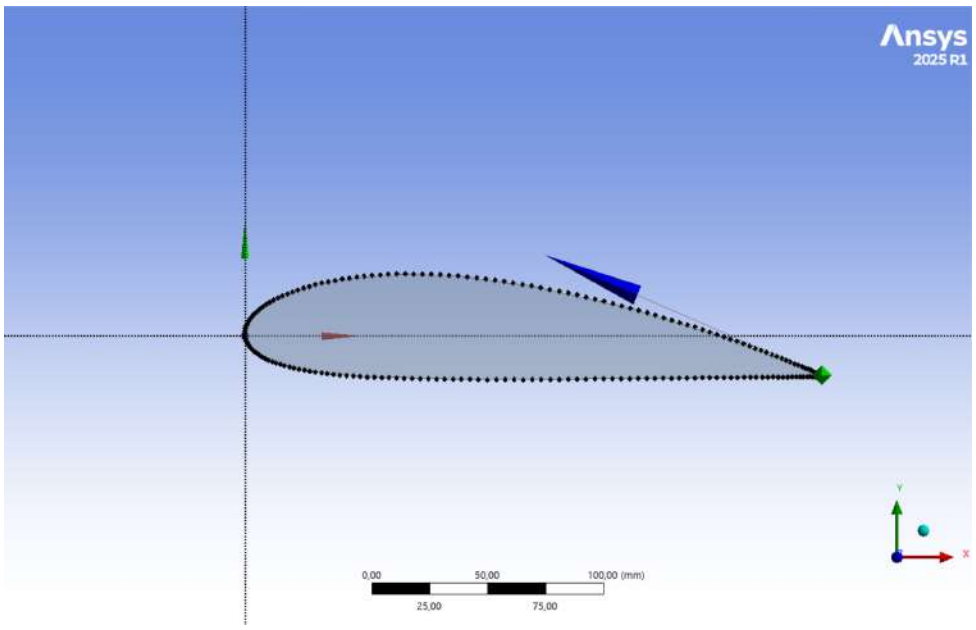


FIGURE 2.2
NACA 4418, 250 mm chord

The propellers are represented by an infinitesimally thin line (shown in green in Figure 2.3). They have a diameter of 130 mm and are positioned below the wing. The propellers are positioned at a distance of 305 mm from the center of mass of the VTOL aircraft. **Note:** From this point forward, the front propeller will be referred to as the left propeller, while the back propeller will be designated as the right propeller.

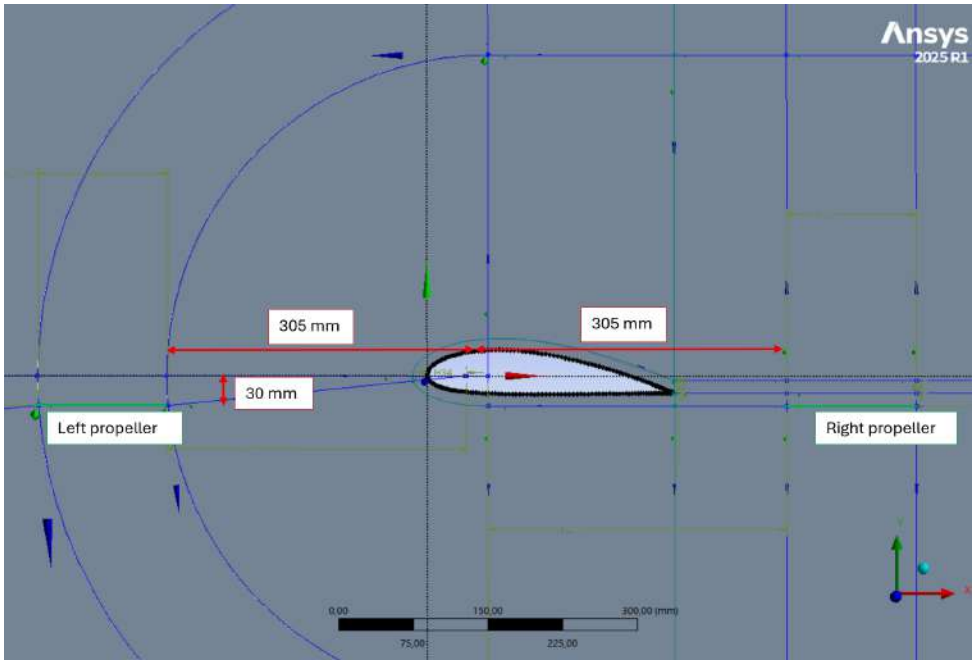


FIGURE 2.3
Emplacement of the propellers.

2.2 UNIT SYSTEM

The International System of Units (SI) is used throughout this study to ensure consistency between the governing equations, material properties and boundary conditions. Using a single, coherent unit system avoids scaling errors and facilitates numerical stability and post-processing, as all quantities are expressed in physically consistent dimensions.

2.3 CHARACTERISTIC DIMENSIONS

The characteristic length used in this study is simplified as the airfoil chord length, as the airfoil is the primary object of interest of the simulation.

However, this simplified choice also represents a limitation of the present approach. The flow configuration under consideration does not correspond to an isolated airfoil, as the presence of propellers both upstream and downstream of the wing strongly modifies the incoming flow conditions and introduces additional characteristic length and velocity scales. As a result, the flow physics is significantly perturbed, and the airfoil chord length alone may not fully capture all relevant aerodynamic interactions in the propeller–wing system.

As mentioned above, the simulations are carried out for four different upstream characteristic velocity values. As a result, the corresponding Reynolds numbers span a range defined by

$$\mathrm{Re} = \frac{Uc}{\nu},$$

where U is the upstream velocity, $c = 0.250\ \mathrm{m}$ is the airfoil chord length, and $\nu = 1.4607 \times 10^{-5}\ \mathrm{m^2/s}$ is the kinematic viscosity of air.

For the lowest velocity case, $U = 3\ \mathrm{m/s}$, the Reynolds number is

$$\mathrm{Re} = \frac{3 \times 0.250}{1.4607 \times 10^{-5}} \approx 5.1 \times 10^4,$$

while for the highest velocity case, $U = 12\ \mathrm{m/s}$, the Reynolds number reaches

$$\mathrm{Re} = \frac{12 \times 0.250}{1.4607 \times 10^{-5}} \approx 2.1 \times 10^5.$$

Within this Reynolds number range, the flow can be confidently assumed to be in the turbulent regime. This assumption is further reinforced by the presence of the propeller-induced flow, which introduces additional disturbances and significantly increases the level of turbulence in the flow field.

2.4 SYMMETRY OF THE PROBLEM

The present study focuses on a two-dimensional cross-section of a single side of the wing of the VTOL RC aircraft. Consequently, once the simulation results are obtained, the contribution of the opposite wing must also be taken into account. Owing to the symmetry of the configuration, the total aerodynamic contribution of the wings can be inferred by assuming identical behavior on both sides.

2.5 GEOMETRICAL SPACE

The simulations are performed using a two-dimensional *planar* formulation. This choice is justified by the assumption that we neglect spanwise variations. Indeed, a planar two-dimensional approach is appropriate

when the geometry, boundary conditions, and flow are assumed to be invariant in the out-of-plane direction. The simplifications resulting from this assumption are twofold. First, for the lift contribution L_1, the induced drag is not taken into account, as this component is inherently three-dimensional and does not appear in a two-dimensional formulation. Second, the propeller disk area is approximated by a square surface with a side length equal to the propeller diameter, as shown in Figure 2.4. This approximation is adopted to simplify the thrust (where T_1 represents left propellers and T_2 represents right propellers) estimation while remaining consistent with the two-dimensional modeling framework.

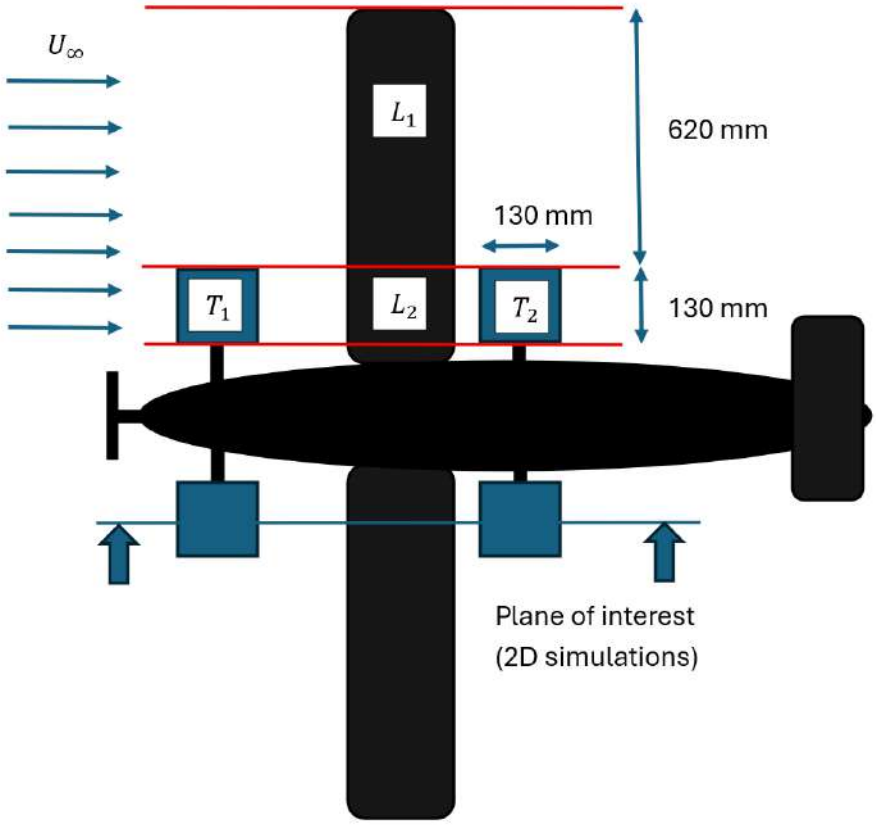


FIGURE 2.4
Top view of the problem under two-dimensional assumptions

This modeling strategy allows the main aerodynamic effects associated with the wing-propeller interaction to be captured while significantly reducing the computational cost.

Nevertheless, this assumption represents a strong simplification and does not fully capture the inherently three-dimensional flow physics of a VTOL configuration, particularly due to the presence of propeller-induced flow structures. A fully three-dimensional simulation would therefore provide a more accurate description of the flow but at the expense of substantially increased computational cost and simulation time.

CHAPTER 3

PHYSICAL MODELING AND HYPOTHESES

3.1 PHYSICAL BEHAVIOR

Air is modeled as a Newtonian fluid in the present simulations. This assumption is justified by the relatively low flow velocities considered, with a maximum upstream velocity of 12 m/s, and by operating conditions close to ambient temperature (288.15 K). Under these conditions, the stress is linearly proportional to the rate of strain.

Furthermore, temperature-dependent viscosity effects are neglected. Given the low Mach number (incompressible regime) of the flow and the absence of significant heat sources, temperature variations within the computational domain are expected to be negligible. As a result, fluid properties are assumed constant, which is consistent with standard CFD practice for low-speed aerodynamic simulations.

3.2 FLUID PROPERTIES

At a temperature of $T = 288.15\ \mathrm{K}$ and under standard atmospheric conditions, air is characterized by a density of $\rho = 1.225\ \mathrm{kg/m^3}$. The dynamic viscosity is $\mu = 1.789 \times 10^{-5}\ \mathrm{Pa \cdot s}$, corresponding to a kinematic viscosity of $\nu = 1.4607 \times 10^{-5}\ \mathrm{m^2/s}$. The specific heat capacity at constant pressure is $c_p = 1005\ \mathrm{J/(kg \cdot K)}$, and the thermal conductivity is $k = 0.025\ \mathrm{W/(m \cdot K)}$. These values are representative of air at ambient conditions and are commonly used in low-speed aerodynamic simulations.

3.3 TURBULENCE MODEL

The **Spalart-Allmaras** turbulence model is used in the present simulations. This model is particularly well suited for external aerodynamic applications involving wall-bounded flows, such as airfoils, where an accurate prediction of boundary-layer behavior is essential. Since the primary objective of this study is the analysis of lift generation on an airfoil during a VTOL transition phase, the Spalart-Allmaras model provides an appropriate compromise between physical accuracy and computational efficiency.

The model is employed with a strain/vorticity-based formulation for the turbulence production term. This choice is especially relevant for the present wing-propeller configuration, as the flow is characterized by strong shear layers, rotational effects, and locally high vorticity levels induced by the interaction between the propeller slipstream and the wing. A strain/vorticity-based formulation enables the turbulence production to better adapt to these shear and rotation-dominated flow features, resulting in a more realistic prediction of turbulence levels in regions directly affected by the propellers.

In addition, the curvature correction option is enabled. This correction is particularly important for airfoil-type geometries, where the flow follows strongly curved streamlines along both the suction and pressure sides of the wing. Curvature effects can significantly influence turbulence production and damping, especially near the leading edge and in regions with strong pressure gradients. Accounting for these effects improves the prediction of boundary-layer development and potential separation behavior.

Although the presence of propellers upstream and downstream of the wing introduces additional flow disturbances, the Spalart-Allmaras model should remain suitable for an accurate analysis of the wing-propeller interaction within a RANS framework. It captures the dominant aerodynamic effects governing lift generation and flow acceleration while avoiding the high computational cost associated with more advanced turbulence models or unsteady simulations.

Concerning the near-wall treatment, the near-wall model approach is employed. This approach consists in resolving the flow field down to the wall, rather than relying on wall functions. As it can be observed in the mesh shown in Figure 3.1, corresponding to the finest grid used in this study, the cells adjacent to the airfoil surface are sufficiently thin to properly resolve the boundary layer. The distance between the two green points indicated in the Figure is on the order of 5×10^{-3} m, which results in y^+ values of approximately 0.06, 0.10, 0.15, and 0.20 for freestream velocities of 3, 6, 9, and 12 $\mathrm{m/s}$, respectively. These values confirm that the viscous sublayer is resolved.

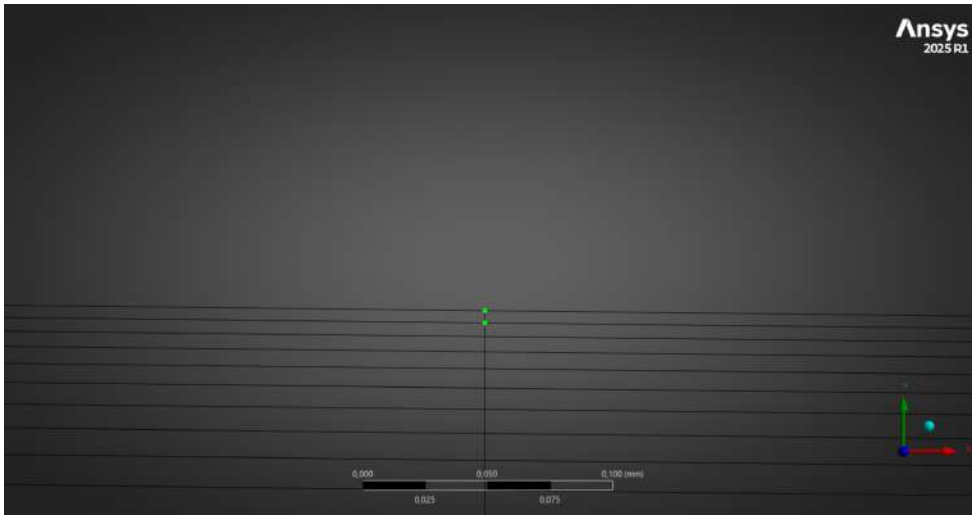


FIGURE 3.1
Close-up view of the finest mesh cell adjacent to the airfoil surface.

This level of mesh refinement allows the near-wall velocity gradients to be accurately captured, which is essential for a reliable prediction of the aerodynamic forces acting on the airfoil.

For a rapid comparison, Figure 3.2 shows the cell size at the airfoil surface for the coarsest mesh used in this study. In this case, the first cell height is on the order of 3×10^{-2} m, which results in y^+ values of approximately 0.35, 0.7, 1.0, and 1.3 for freestream velocities of 3, 6, 9, and 12 $\mathrm{m/s}$, respectively. These values remain acceptable for a coarse mesh and are still consistent with a near-wall modeling approach.

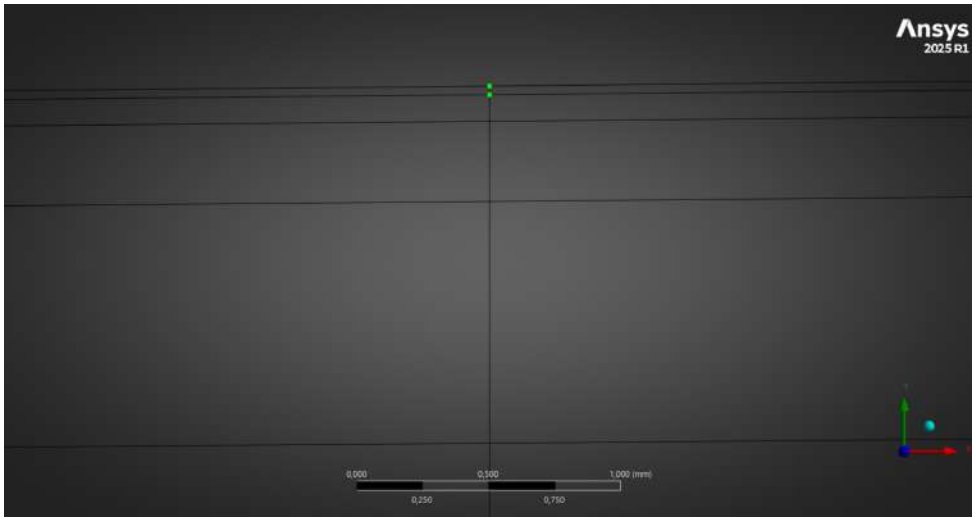


FIGURE 3.2
Close-up view of the coarsest mesh cell adjacent to the airfoil surface.

Chapter 4

Boundary conditions, external forces, initial conditions

4.1 Boundary conditions

4.1.1 Inlet

In Figure 4.1, the inlet boundary is shown geometrically by the green line. In *ANSYS Fluent*, this boundary is defined using a velocity inlet boundary condition. Since the airfoil already has an angle of attack of $4°$, the inlet velocity vector is prescribed with components $(1, 0)$ for all imposed velocity magnitudes.

The gauge pressure at the inlet is set to 0 Pa, ensuring that the inlet pressure is equal to the operating pressure and the turbulence specification method is the turbulent viscosity ratio, which is set to a value of 10.

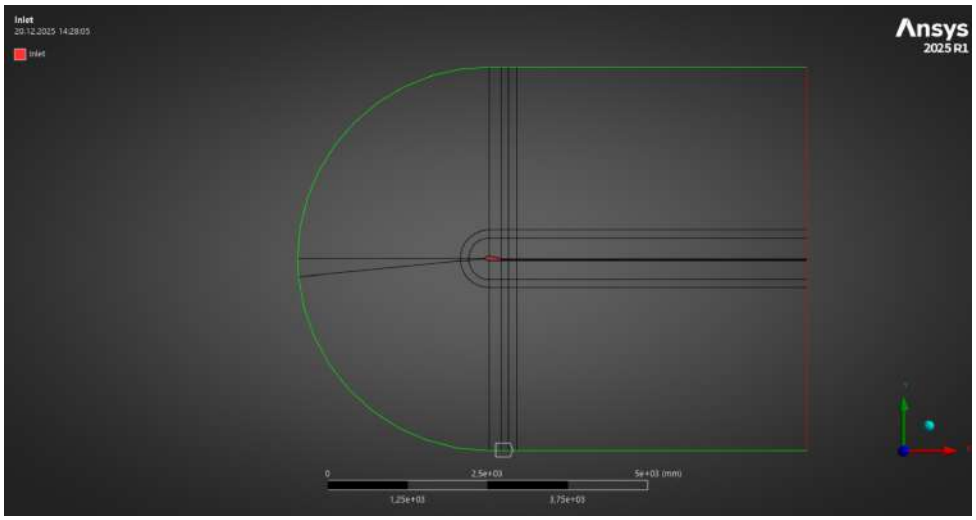


Figure 4.1
Inlet boundary.

4.1.2 Outlet

In Figure 4.2, the outlet boundary is represented geometrically by the green line. In *ANSYS Fluent*, this boundary is implemented using a pressure-outlet boundary condition.

The gauge pressure at the outlet is also set to 0 Pa, and the turbulent viscosity ratio is prescribed with the same value as at the inlet.

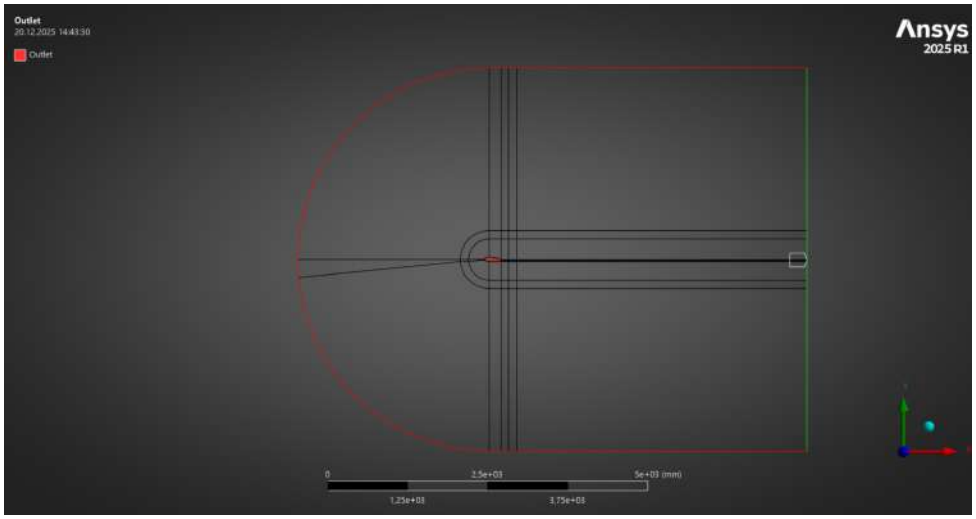


FIGURE 4.2
Outlet boundary.

4.1.3 AIRFOIL SURFACE

In Figure 4.3, the airfoil surface boundary is represented geometrically by the green line. In *ANSYS Fluent*, this boundary is implemented as a wall boundary condition with a no-slip condition applied.

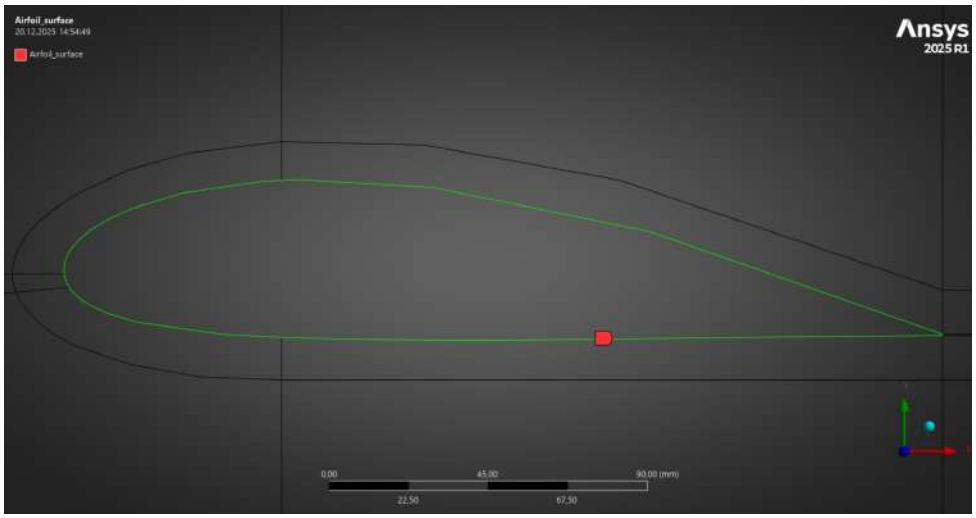


FIGURE 4.3
Airfoil boundary.

4.1.4 PROPELLERS

In Figure 4.6, the propeller boundary is geometrically represented by the green lines. The boundary condition is implemented using the fan boundary condition available in ANSYS Fluent. To be applied, this model requires a one-dimensional element (a straight line) representing the propellers. The boundary condition then imposes a pressure difference between the two sides of this one-dimensional element.

To better capture the fan dynamics, the polynomial mode was selected. Instead of imposing a constant pressure difference regardless of the environmental conditions, this mode defines a relationship between the fan pressure rise and the normal component of the velocity across the fan.

To define this relationship, an experimental equation commonly found in the literature[1] was used to estimate the thrust produced by a propeller at a given rotational speed (RPM):

$$F = 4.392399 \times 10^{-8}\,\mathrm{RPM}\,\frac{d^{3.5}}{\sqrt{\mathrm{pitch}}}\left(4.23333 \times 10^{-4}\,\mathrm{RPM}\cdot\mathrm{pitch} - V_n\right) \tag{4.1}$$

By rearranging this equation, a relationship of the form $\Delta P(V_n)$ can be derived:

$$\Delta P(V_n) = \frac{4.392399 \times 10^{-8}\,\mathrm{RPM}\,\dfrac{d^{3.5}}{\sqrt{\mathrm{pitch}}}}{A}\left(4.23333 \times 10^{-4}\,\mathrm{RPM}\cdot\mathrm{pitch} - V_n\right) \tag{4.2}$$

This expression defines a linear relationship between the pressure rise across the fan and the normal velocity component, as illustrated in Figure 4.4 for several throttle percentages. This behavior is consistent with the polynomial boundary condition fan model implemented in ANSYS Fluent.

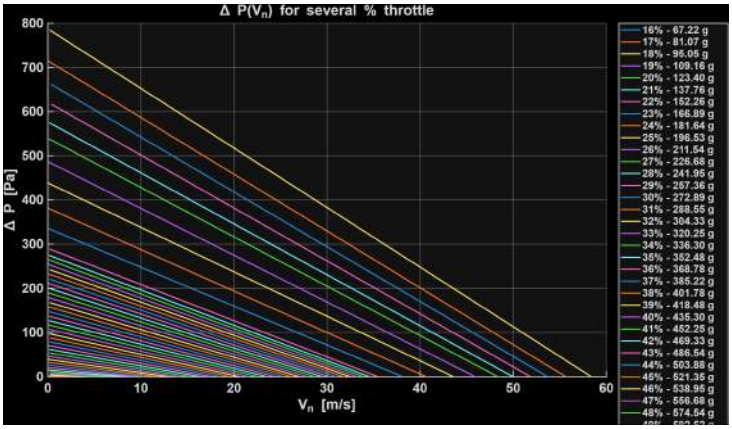
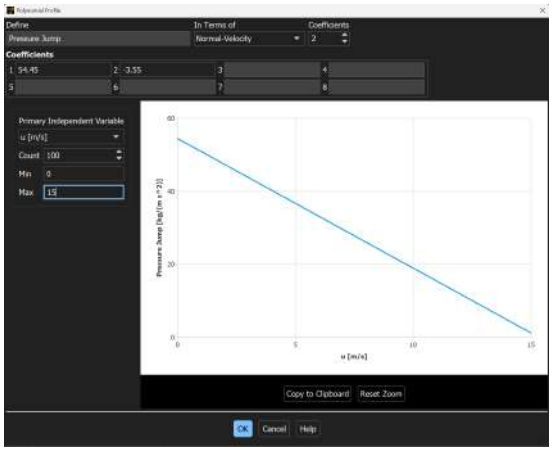


FIGURE 4.4
Pressure jump in function of the V_n normal velocity to the propeller edge boundary condition.

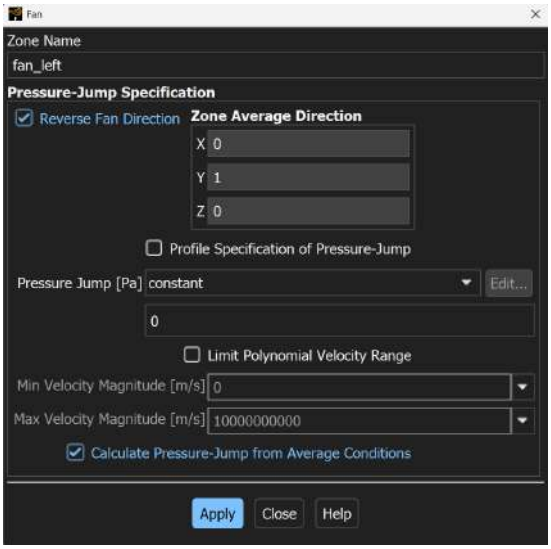
For a given throttle setting, the rotational speed is obtained from a polynomial interpolation of the experimental RPM–throttle data found in the propeller's datasheet (AliExpress 2024). The static thrust corresponding to this operating condition is then used to determine the maximum pressure rise at zero normal velocity, ensuring consistency with the measured propeller performance.

Finally, the resulting $\Delta P(V_n)$ relationship is approximated by a first-order polynomial and directly implemented in the Fluent fan boundary condition as shown in Figure 4.5 (A). Furthermore, the pressure jump obtained from the simulations is computed using averaged flow conditions at the edge cells defining the propeller boundary conditions, shown in Figure 4.5 (B).

[1]This empirical equation was obtained from an online reference presenting experimental correlations for propeller thrust:`https://www.electricrcaircraftguy.com/2014/04/propeller-static-dynamic-thrust-equation-background.html`



(A) The pressure-jump polynomial used to model the propellers physic (shown here for a propeller throttle setting of 27%).



(B) Fan boundary condition set up.

FIGURE 4.5

Given the limited operating range investigated and the absence of a complete experimental fan map, this linear approximation was considered sufficient to capture the dominant behavior of the propeller. It should be noted that this modeling approach neglects blade-induced swirl effects and higher-order nonlinearities, which are implicitly lumped into the experimentally derived coefficients.

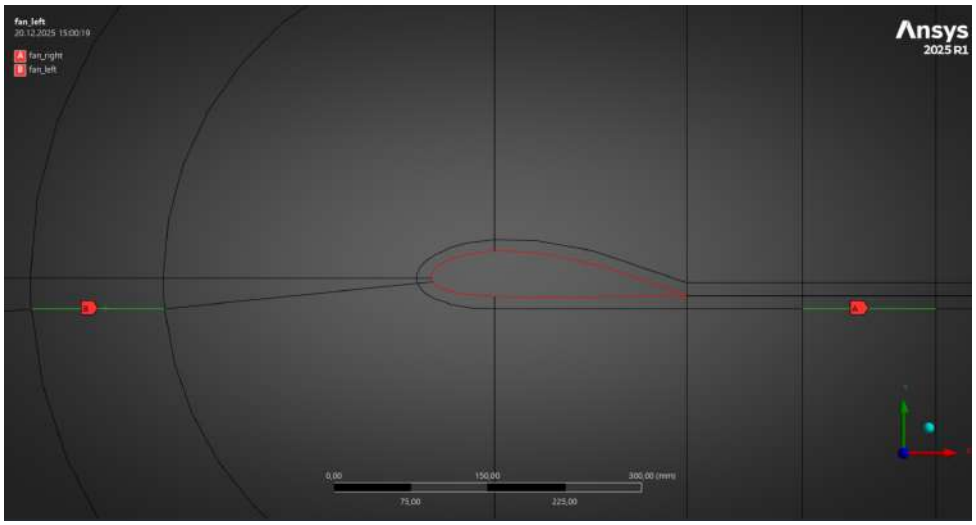


FIGURE 4.6
Propellers boundary.

Chapter 5

Computational mesh

5.1 Mesh type

A quadrilateral structured mesh is used in this CFD study. The mesh is generated using the sizing tools available in ANSYS Workbench Mechanical, where each edge is assigned a prescribed number of divisions in order to obtain a structured mesh topology. This choice is motivated by the relatively simple geometry of the airfoil cross-section and by the need to accurately resolve the boundary layer. Structured meshes provide good control over cell distribution, alignment, and smoothness, which helps reduce numerical diffusion and improves the accuracy of gradient and flux computations, particularly in wall-bounded aerodynamic flows.

5.2 Cell type

A face meshing approach using a quadrilateral method is applied to all faces throughout the computational domain. This choice is well suited for the present two-dimensional aerodynamic simulation. Quadrilateral elements offer improved control over mesh quality metrics such as orthogonality, smoothness and aspect ratio, which directly influence the accuracy and stability of the numerical solution. These properties are particularly important in turbulent flows, where strong velocity gradients and complex flow structures are present.

Moreover, the use of quadrilateral cells enables the generation of a smooth and structured mesh, allowing for a gradual refinement toward the airfoil surface. This is especially beneficial for resolving the boundary layer and capturing near-wall flow behavior accurately, even in the presence of flow disturbances induced by the propellers. Quadrilateral elements are therefore a preferred choice for structured meshes in external aerodynamic flows, such as airfoil simulations.

Nevertheless, it should be noted that the use of quadrilateral elements does not eliminate all numerical errors, and mesh quality must still be carefully monitored in regions with strong flow disturbances and high gradients.

5.3 Size and number of cells of the finest mesh

The finest mesh used in this study contains $411{,}225$ nodes and $410{,}096$ elements. Local mesh refinements are applied in the vicinity of the airfoil and the propellers in order to accurately capture the strong gradients and flow features in these regions as already mentioned.

5.3.1 AIRFOIL EDGE SIZING

The element length along the airfoil surface is set to $1.45\ \mathrm{mm}$. In addition, a bias factor of 250 (with a cell growth rate set to 1.2) is applied in the normal direction to the airfoil surface in order to achieve sufficient mesh refinement near the wall as shown in Figure 5.1. This strong grading allows the near-wall region to be properly resolved, which is required for the near-wall modeling approach described previously.

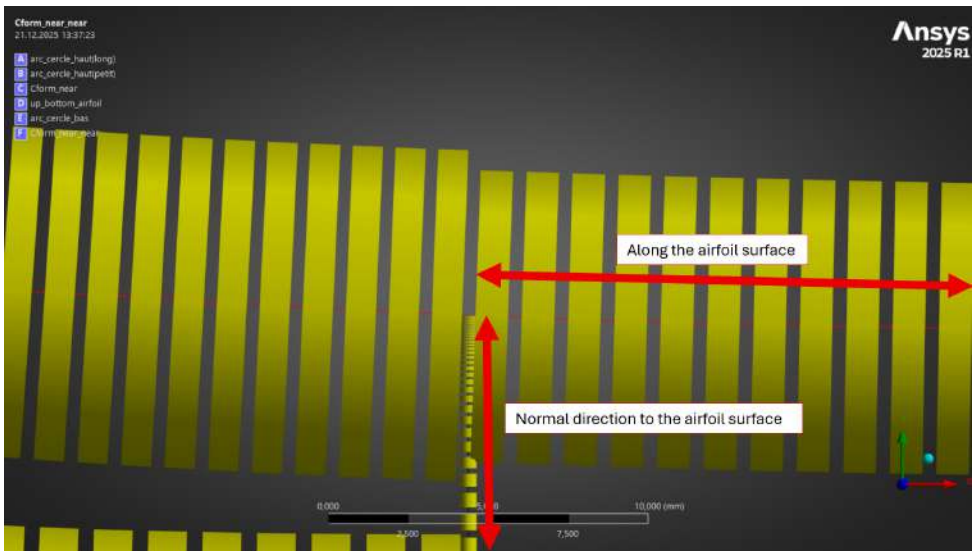


FIGURE 5.1
Mesh refinement at the airfoil edge.

5.3.2 PROPELLERS EDGE SIZING

For the propellers, the element length along their span is chosen to be identical to that used on the airfoil surface, namely $1.45\ \mathrm{mm}$, as illustrated in Figure 5.2.

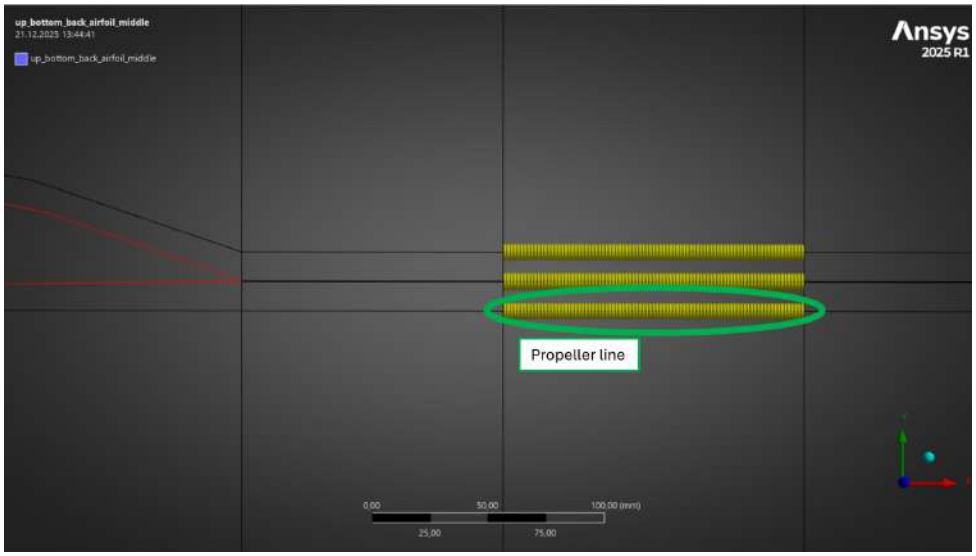


FIGURE 5.2
Mesh refinement along the right propeller's edge.

5.3.3 WAKE EDGE SIZING

The wake region, downstream the right propeller, is meshed using a bias factor of 128 in order to progressively increase the cell size downstream, leading to a coarser mesh in regions where high spatial resolution is no longer required. As shown in Figure 5.3, the transition between fine and coarse cells is very smooth. According to standard meshing guidelines, such a gradual variation in cell size helps limit truncation errors, preserves the accuracy of second-order discretization schemes, and improves numerical stability by avoiding abrupt changes in mesh resolution.

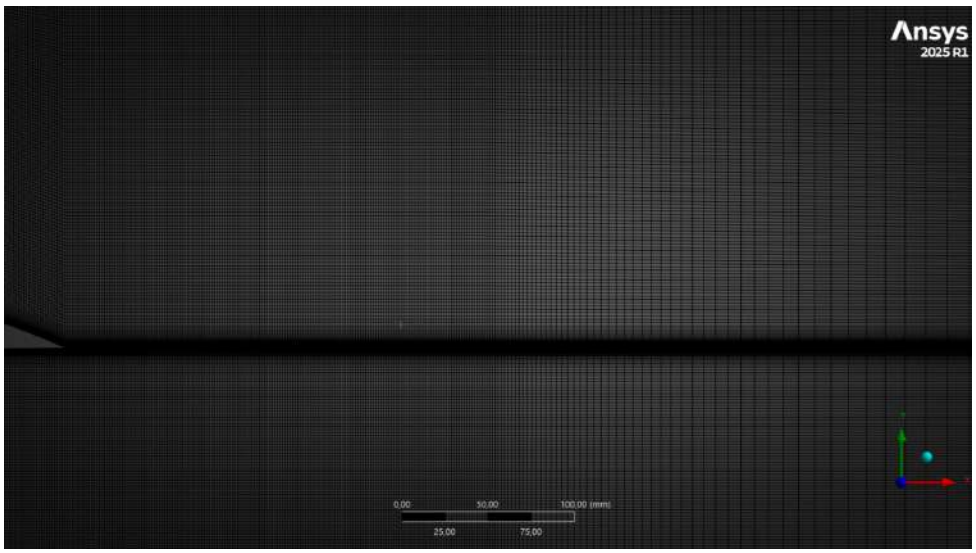
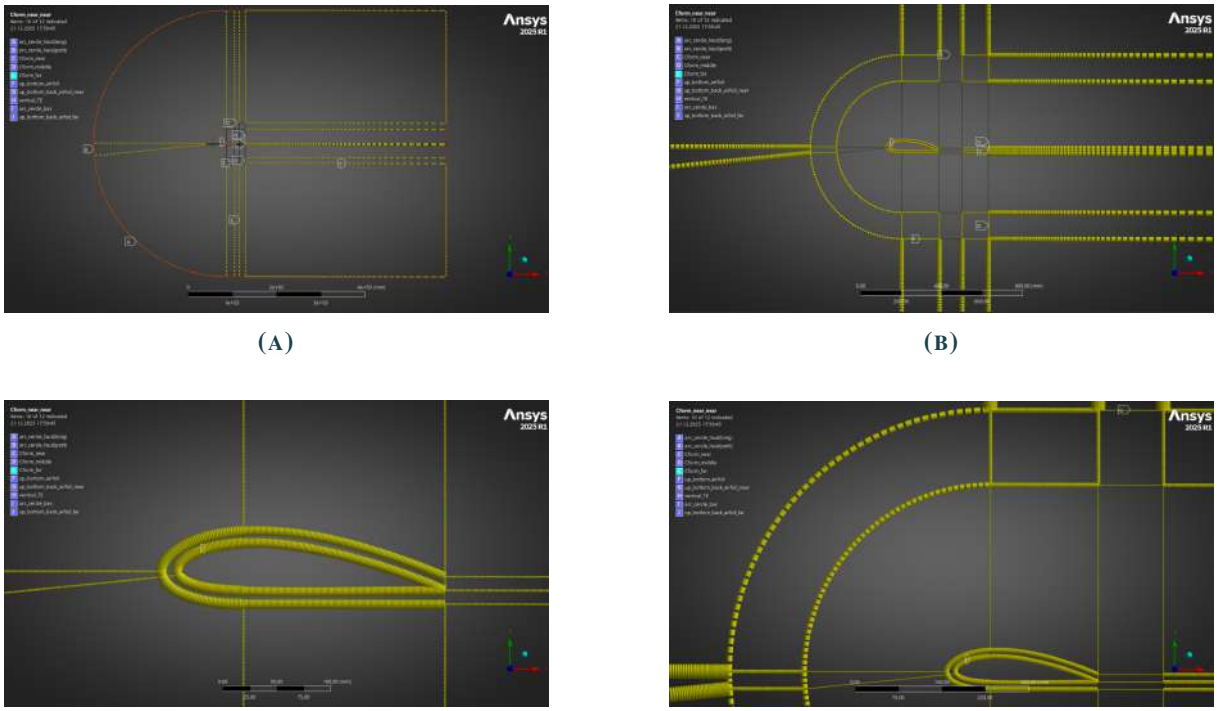


FIGURE 5.3
Smooth transition of the mesh in the wake region.

5.4 FINEST MESH PRESENTATION

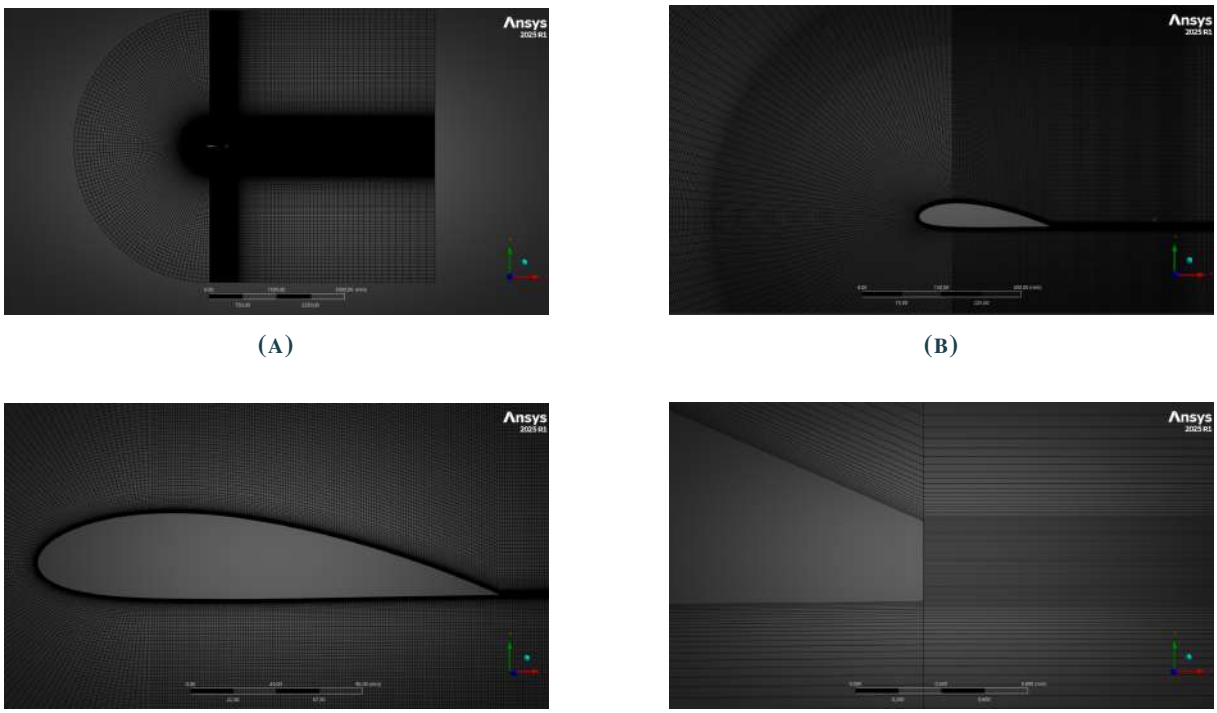
In Figure 5.4, the different edges and the number of divisions assigned to each of them are shown, forming the structural "skeleton" of the mesh. The resulting mesh generated from this edge discretization is shown in Figure 5.5. All details regarding the number of edge divisions used to generate the finest mesh are provided in the second appendix.



(A) (B)

(C) (D)

FIGURE 5.4
Global view and detailed close-ups of the edge sizing applied to the geometry.



(A) (B)

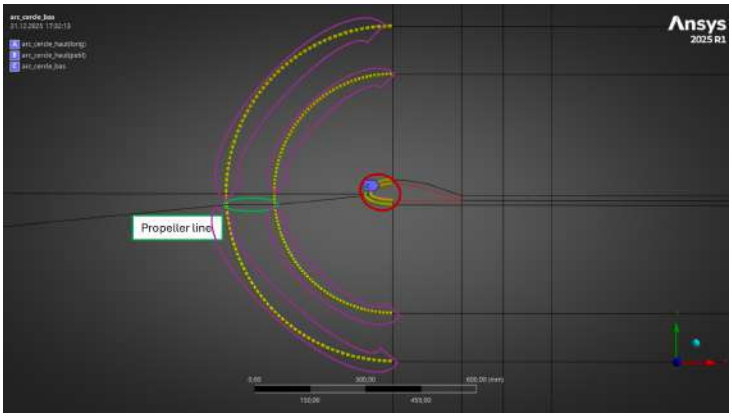
(C) (D)

FIGURE 5.5
Global view and detailed close-ups of the mesh.

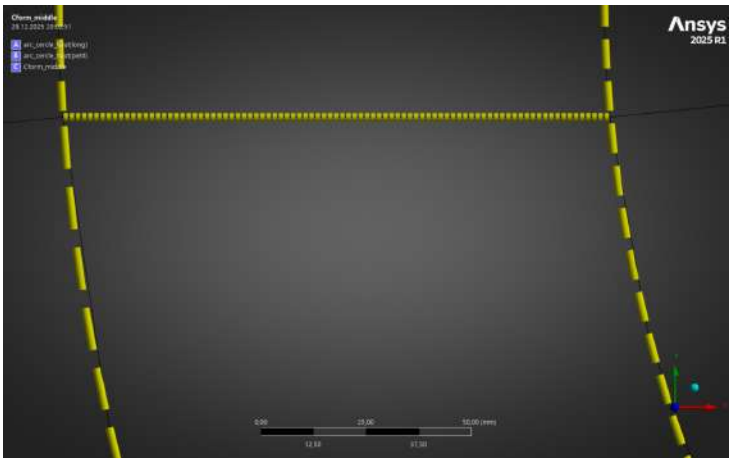
Although the regular structure of the mesh appears satisfactory at first glance, two major issues arise from the use of this structured mesh. These issues are discussed in detail in the following points A and B.

A. LIMITATIONS OF THE STRUCTURED MESH NEAR THE NORMAL DIRECTION OF THE LEFT PROPELLER

In order to incorporate the propeller lines within a structured mesh, several compromises had to be made. Specifically, when a given number of divisions is imposed on the leading edge of the airfoil (red circle in Figure 5.6 (A)), the opposite edges must have the same number of divisions to preserve the structured topology. As a consequence, the regions located above and below the left propeller (purple regions in Figure 5.6 (A)) are meshed with a lower spatial resolution in the normal direction well visible in Figure 5.6 (B). Indeed, while the prescribed number of divisions allows for small elements along the relatively short airfoil edge length, applying the same division count over a larger geometric length inevitably results in larger mesh elements in those regions.



(A) Overview of the edge sizing applied to the leading edge and the opposite edges.



(B) Zoomed view on the edge sizing around the propeller line

FIGURE 5.6

This issue does not affect the right propeller, which retains a sufficiently fine resolution in the direction normal to its edge (red region in Figure 5.7)

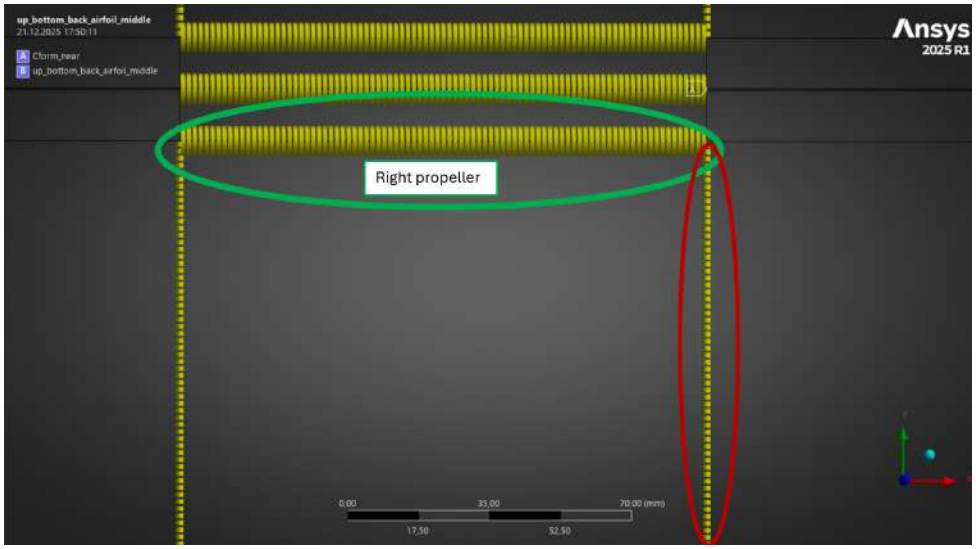


FIGURE 5.7

.

B. ABRUPT CELL SIZE VARIATIONS INDUCED BY A STRUCTURED MESH

An abrupt change in cell size can be observed in the red region of Figure 5.8. This issue arises because the edge sizing applied to the upstream circular arc of the airfoil, shown in Figure 10.11 in the appendix, differs from the sizing imposed on the edges located above and below the airfoil, shown in Figure 10.8 in the appendix. As a result, a significant cell size discontinuity occurs at their intersection.

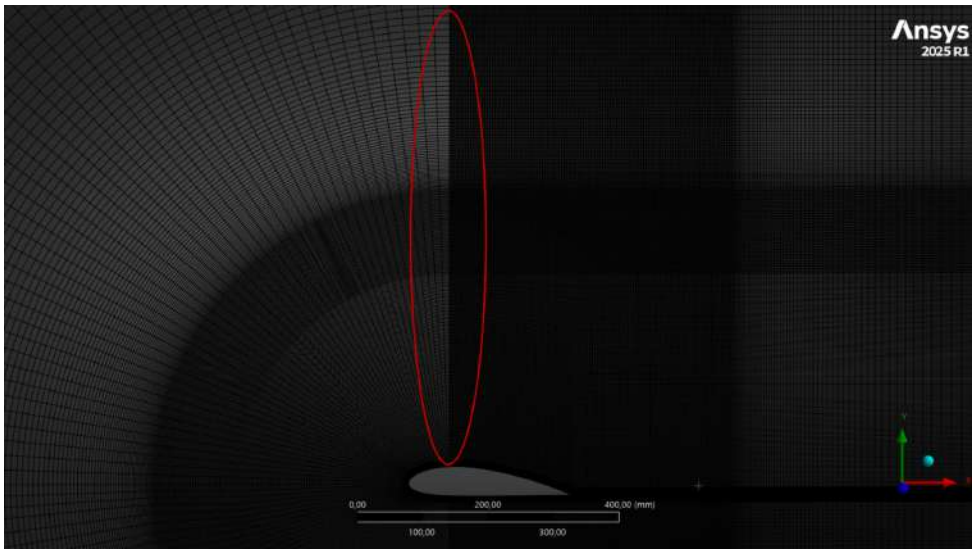


FIGURE 5.8
Abrupt jump in cell size.

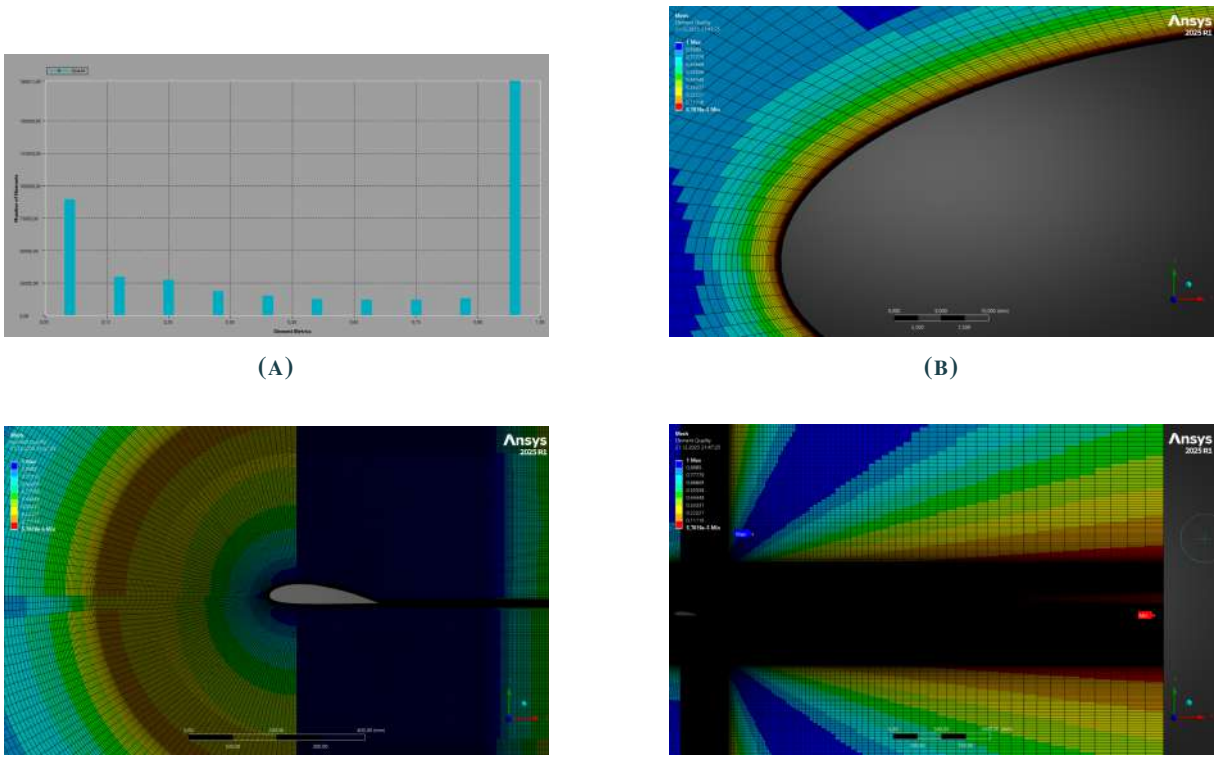
This issue progressively vanishes when approaching the airfoil, which mitigates the numerical errors associated with the abrupt cell size variation. According to meshing best practices, errors induced by cell size jumps are most critical in regions with strong gradients. Since the mesh is significantly refined in the vicinity of the airfoil (where the dominant aerodynamic gradients are located) the impact of this size discontinuity on the overall solution accuracy is reduced.

5.5 QUALITY METRICS

5.5.1 ELEMENT QUALITY

As shown in Figure 5.9 (A), the overall mesh quality is considered satisfactory. Regions exhibiting low element quality values close to zero are localized and occur in areas where such quality degradation is acceptable from a numerical standpoint. In particular, near the airfoil surface, as illustrated in Figure 5.9 (B), the low-quality elements correspond to highly stretched cells within the boundary layer. This stretching is intentional and aligned with the direction of the strongest velocity gradients, which occur normal to the wall. According to meshing guidelines, such anisotropic elements are appropriate for resolving boundary layers, provided that the cell orthogonality and near-wall resolution are maintained.

Similarly, in the wake region downstream of the airfoil, as shown in Figure 5.9 (D), low-quality values are associated with elongated cells aligned with the main flow direction. In this region, flow gradients are predominantly streamwise, and cell elongation in the direction of the flow helps reduce numerical diffusion while maintaining computational efficiency. Low element quality values are not inherently problematic when they occur in regions of smooth flow variation and when the cell stretching is physically consistent with the dominant flow features.



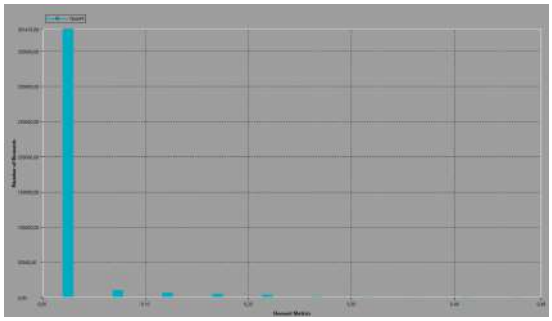
(A) (B) (C) (D)

FIGURE 5.9
Element quality metric.

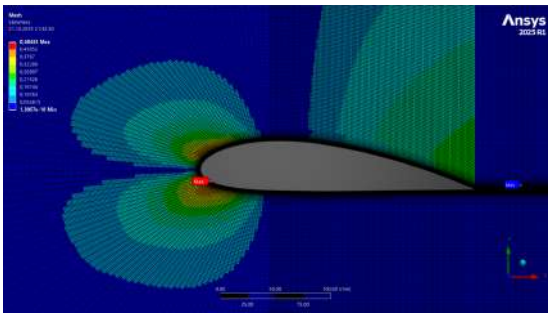
5.5.2 SKEWNESS

As shown in Figure 5.10 (A), the overall skewness of the mesh is very good. The vast majority of elements exhibit low skewness values, which indicates well-shaped cells and is favorable for numerical accuracy and solver robustness. A small number of cells located near the leading edge, Figure 5.10 (B) and (C), of the airfoil present skewness values slightly below 0.5. Although this is not ideal, these values remain well below the critical threshold of 1.

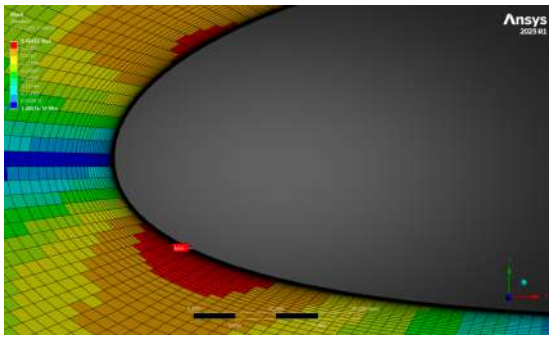
Skewness values significantly lower than 1 are generally acceptable and are not expected to cause convergence issues. Moreover, the cells exhibiting higher skewness are confined to a very limited region and coincide with strong geometric curvature at the leading edge. Localized skewness in such regions is often unavoidable in structured meshes and does not significantly affect solution stability. Therefore, the observed skewness distribution is not expected to negatively impact the convergence or accuracy of the simulations.



(A)



(B)

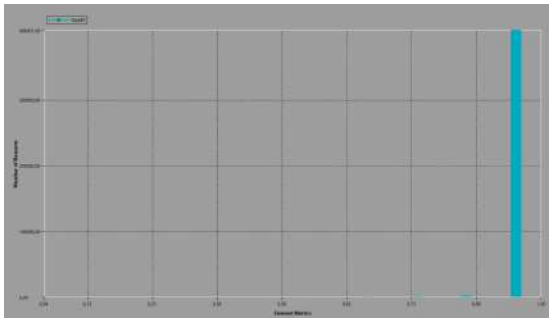


(C)

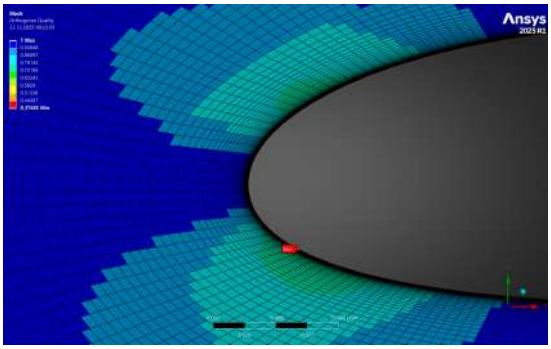
FIGURE 5.10
Skewness metric.

5.5.3 ORTHOGONALITY

As shown in Figure 5.11 (A), the overall orthogonality of the mesh is excellent, with most elements exhibiting high values that are favorable for accurate gradient and flux computations and for numerical stability. Local orthogonality values around 0.35 are observed near the leading edge of the airfoil, as illustrated in Figure 5.11 (B), in the same region where slightly higher skewness values were previously identified. This local degradation is primarily due to the strong geometric curvature at the leading edge, which inevitably affects cell alignment in structured meshes. Since this reduction in orthogonality is spatially limited and occurs in a highly refined region, it is not expected to significantly affect the stability or convergence of the simulations.



(A)



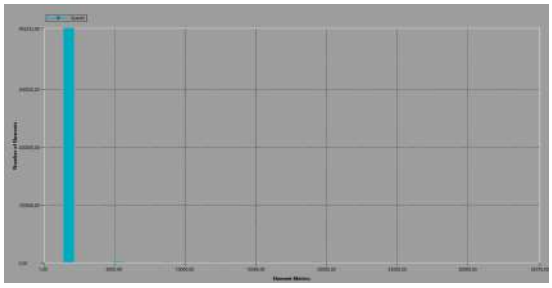
(B)

FIGURE 5.11
Orthogonality metric.

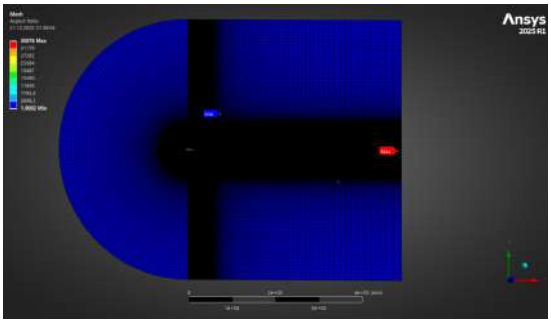
5.5.4 ASPECT RATIO

As shown in Figure 5.12 (A), the overall aspect ratio of the mesh is very good. High aspect ratio cells are mainly located in the wake of the airfoil, as illustrated in Figure 5.12 (C).

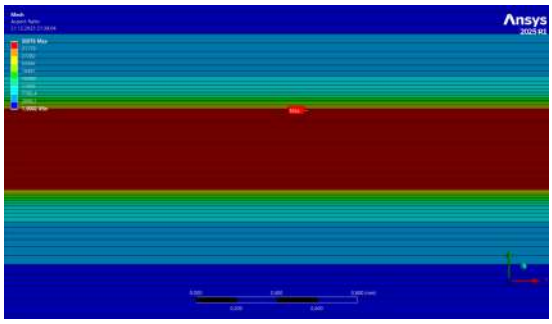
High aspect ratio elements are acceptable when they are intentionally introduced and aligned with the main flow direction. In the wake region, they allow the gradual downstream evolution of the flow to be captured with a reduced computational cost. The largest aspect ratio values are observed far downstream of the airfoil, where flow gradients are weak, and the solution is less sensitive to local mesh quality. Consequently, the observed aspect ratio distribution is not expected to adversely affect convergence or numerical stability.



(A)



(B)



(C)

FIGURE 5.12
Aspect ratio metric.

CHAPTER 6

NUMERICAL METHODS

6.1 SPATIAL DISCRETIZATION

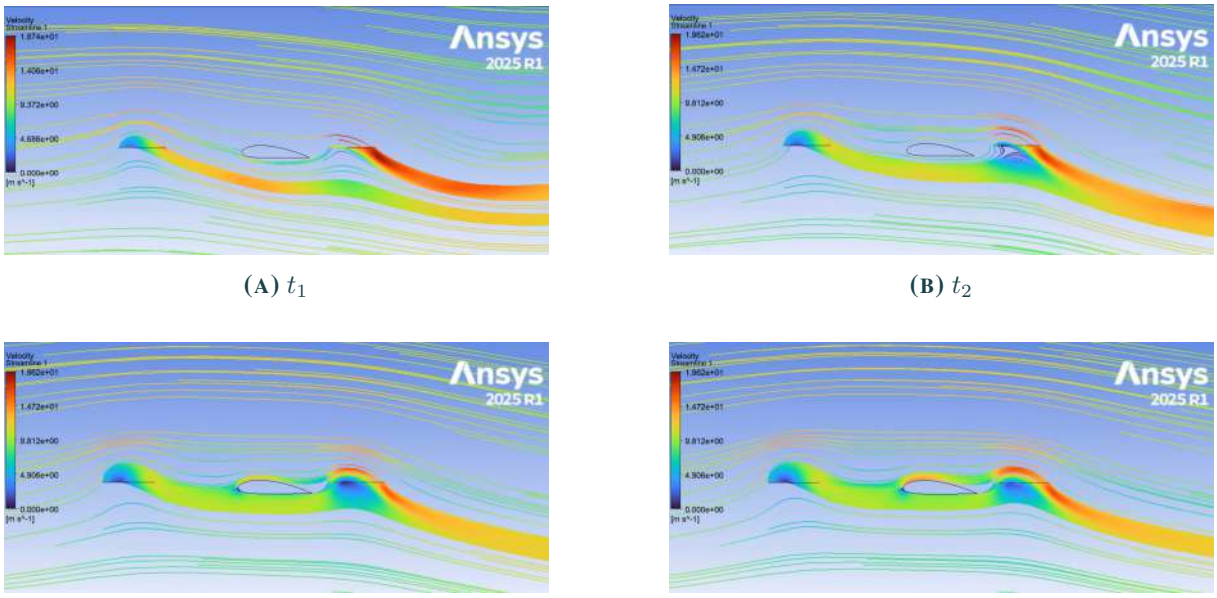
The simulations are performed using the finite volume method (FVM), which is the numerical framework implemented in *ANSYS Fluent*. In this approach, the governing equations are integrated over control volumes, ensuring strict conservation of mass, momentum, and turbulence quantities. This property is particularly well suited for aerodynamic simulations involving complex flow features such as boundary layers and wakes, as encountered in the present wing–propeller configuration.

Second-order spatial discretization schemes are employed for pressure, momentum, and the modified turbulent viscosity equation of the Spalart-Allmaras model. Second-order schemes significantly reduce numerical diffusion compared to first-order schemes, allowing for a more accurate representation of velocity gradients, pressure distributions, and wake development. This is especially important for lift prediction and for capturing the interaction between the airfoil and the propeller-induced flow.

The main drawback of second-order discretization is a reduced numerical robustness compared to first-order schemes. Second-order schemes are more sensitive to mesh quality and may lead to convergence difficulties if strong gradients, poor cell quality or inadequate initial conditions are present. However, given the relatively good overall mesh quality, second-order accuracy is considered an appropriate compromise between accuracy and numerical stability for the present study.

6.2 TYPE OF SIMULATION AND SOLVER USED

At first glance, the superposition of a horizontal incoming flow with the vertical momentum induced by the propeller pressure jump could suggest a highly unsteady and chaotic flow field, for which a transient approach would be required to correctly capture the flow physics. However, transient simulations performed with a time step of $0.005\ \mathrm{s}$, 1500 time steps and 20 iterations per time step were also carried out to assess the unsteady behavior of the flow. With an upstream velocity of $8\ \mathrm{m/s}$ and the propellers operating at 50% of their maximum power show that the flow progressively stabilizes in time. After an initial transient phase, the flow reaches a quasi-steady state that can reasonably be considered steady, as illustrated in Figure 6.1.



(A) t_1 (B) t_2

(C) t_3 (D) t_4

FIGURE 6.1
Illustration of the flow converging toward a steady state

This observation allows us to subsequently assume that the simulations can be performed under steady-state conditions in certain cases. As will be shown in the results section, for certain combinations of upstream velocity U and propeller throttle percentage, this assumption may be somewhat strong. Nevertheless, steady-state simulations remain capable of capturing the main characteristics of the flow field. In addition, in cases where the residuals do not converge below the prescribed threshold, time-averaged quantities and the associated standard deviation are computed to ensure a meaningful interpretation of the results, as it will be explained in more details in Section 6.3.

Furthermore, given the methodology adopted in this study (namely determining, for each upstream velocity U, the propeller throttle percentage required to satisfy the force balance) fully transient simulations would be extremely time-consuming. In particular, accurately capturing the flow physics requires the use of a Courant number in the range of 0.5 to 1, which significantly increases the time computational cost. Consequently, the steady-state assumption represents the best compromise between simulation time and accuracy for the objectives of the present analysis.

In the present analysis, a pressure-based solver is employed. This choice is well suited for incompressible and low-Mach-number flows, where pressure–velocity coupling plays a central role in enforcing mass conservation. These solvers are particularly appropriate for external aerodynamic applications involving moderate velocities, such as the present wing–propeller configuration.

In addition, a coupled pressure–velocity scheme is selected because the flow is characterized by strong interactions between pressure and velocity fields. The pressure jump imposed by the propellers directly affects the local velocity field, while the resulting velocity variations modify the pressure distribution around the airfoil. Solving pressure and velocity simultaneously should therefore improve convergence robustness and numerical stability.

In the present simulations, the flow is modeled as incompressible, isothermal and chemically non-reacting. As a result, the primary variables solved for are the velocity components and the pressure field. In addition, one turbulence-related variable, namely the modified turbulent viscosity $\tilde{\nu}$, is solved as part of the Spalart-Allmaras turbulence model. The density is assumed constant (Mach « 0.3) and is therefore

not solved for, while the temperature equation is not activated since thermal effects are neglected. No transport equations for chemical species are considered, as the flow involves air only and no chemical reactions. Consequently, the governing equations consist of the incompressible continuity equation, the momentum equations, and the transport equation for the Spalart-Allmaras turbulence variable.

CONTINUITY EQUATION

$$\nabla \cdot \mathbf{u} = 0 \tag{6.1}$$

where $\mathbf{u}$ denotes the mean velocity vector. This equation enforces mass conservation for an incompressible flow.

MOMENTUM EQUATIONS

$$\rho\left(\mathbf{u} \cdot \nabla \mathbf{u}\right) = -\nabla p + \nabla \cdot \left[\left(\mu + \mu_t\right)\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)\right] \tag{6.2}$$

where ρ is the fluid density, p is the mean pressure, μ is the molecular dynamic viscosity, and μ_t is the turbulent (eddy) viscosity introduced by the turbulence model. The term $\nabla \mathbf{u}^T$ denotes the transpose of the velocity gradient tensor. The additional viscosity μ_t accounts for the Reynolds stresses arising from turbulence.

SPALART-ALLMARAS TURBULENCE MODEL

$$\mathbf{u} \cdot \nabla \tilde{\nu} = C_{b1}\left(1 - f_{t2}\right)\tilde{S}\tilde{\nu} + \frac{1}{\sigma}\left[\nabla \cdot \left((\nu + \tilde{\nu})\nabla \tilde{\nu}\right) + C_{b2}(\nabla \tilde{\nu})^2\right] - C_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2 \tag{6.3}$$

where $\tilde{\nu}$ is the modified turbulent kinematic viscosity, $\nu = \mu/\rho$ is the molecular kinematic viscosity, $\tilde{S}$ is a modified measure of the strain-rate magnitude accounting for rotation effects, and d is the distance to the closest wall. The coefficients C_{b1}, C_{b2}, C_{w1}, and σ are empirical model constants, while f_{t2} and f_w are damping functions used to model near-wall and turbulence destruction effects. The turbulent viscosity entering the momentum equations is obtained from $\tilde{\nu}$ through an algebraic relation.

6.3 SOLUTION OPTIONS

The simulations are initialized using the hybrid initialization method. This approach combines a potential-flow-based initialization with local corrections to provide a physically consistent initial flow field throughout the computational domain. Compared to constant or piecewise constant initialization, hybrid initialization significantly improves robustness and accelerates convergence, particularly for complex configurations involving multiple boundary conditions, such as the present wing-propeller interaction. It provides a better initial guess for velocity and pressure fields without requiring a previously converged solution.

All simulations are performed under steady-state conditions. Convergence is primarily assessed based on residual reduction, with convergence considered achieved when the residuals of the continuity equation, the x- and y-momentum equations and the modified turbulent viscosity equation ($\tilde{\nu}$) of the Spalart-Allmaras model drop below 10^{-6}. In addition, a maximum number of 2000 iterations is imposed. In practice, this iteration limit often governs the termination of the simulations, as residuals may exhibit oscillatory behavior due to the strongly coupled wing–propeller flow. In such cases, the solution is still considered acceptable, as the oscillations remain bounded and the flow quantities of interest reach statistically stable values. When residual convergence is not strictly achieved and the maximum number of 2000 iterations is reached, statistical averaging of the relevant quantities, such as the wing lift and

the propeller's thrust, is performed over the final 500 iterations. In addition, the corresponding standard deviation is computed in order to provide a critical assessment of the obtained values and to quantify the level of residual oscillations.

6.4 COMPUTED QUANTITIES

For the propellers, at each iteration, the mean pressure jump ΔP [Pa] is computed across the interface representing the front and rear propeller boundary condition. When residuals convergence is not strictly achieved, this quantity is first time-averaged over the final iterations, as explained before. Then, the thrust generated by each propeller is computed as

$$T_{\text{left prop}} = T_1 \approx \Delta P_{\text{left prop}} \times 0.13^2 \times 2,$$

$$T_{\text{right prop}} = T_2 \approx \Delta P_{\text{right prop}} \times 0.13^2 \times 2.$$

Regarding the lift, the contribution of the wing portion disturbed by the propellers is obtained from the sectional lift per unit span, denoted $L_{2,b}$ [N/m] computed at each iteration. The total lift contribution is computed as

$$L_2 = L_{2,b} \times 2 \times 0.13.$$

Finally, it is recalled that the lift contribution L_1, corresponding to the undisturbed portion of the wing, is computed separately using an independent two-dimensional simulation with a simple and structured mesh closely resembling that employed in the present study.

Chapter 7

Results

7.1 Overview

This chapter presents the simulation results for four different cruise speeds performed on the finest mesh. For each cruise speed, various throttle inputs were analyzed. Given the time constraints of the project, it was not feasible to explore the entire range of possible throttle inputs and boundary conditions. Therefore, an iterative approach was adopted: starting with an initial throttle estimate, the resulting lift was evaluated, and the throttle was adjusted iteratively until a comprehensive overview was obtained for each cruise velocity. The primary objective is to achieve a weight-to-lift ratio as close to unity as possible, which would indicate that the drone is capable of maintaining altitude during the transition from vertical to horizontal cruise.

7.2 Key Observations

7.2.1 Lift Behavior Across Velocity Range

The results presented in Tables 7.1–7.4 reveal several important trends across the velocity spectrum. In the low velocity regime, specifically at 3 m/s and 6 m/s, lift increases monotonically with increasing throttle percentage. However, a weight-to-lift ratio of unity is never achieved within the tested throttle range, indicating that propeller thrust alone is insufficient to maintain altitude at these low forward speeds. This limitation suggests that the drone cannot sustain level flight during the initial phase of the transition without losing altitude.

Conversely, at higher velocities around 9 m/s, maximum lift occurs at partial throttle settings rather than at maximum power. Further increases in throttle percentage beyond this optimum result in a clear reduction in total lift. This behavior suggests an adverse interaction between the propeller-induced flow and the wing aerodynamics at high throttle settings, where the propeller wake may disrupt the favorable pressure distribution over the wing surface.

The hovering condition at 0 m/s indicates a required throttle of approximately 50%, which is consistent with standard values reported in the literature for hovering flight. This observation confirms that the propellers are adequately sized for this application and validates the propeller selection process. At the opposite end of the velocity spectrum, the optimal cruise condition at 12 m/s with zero throttle input achieves a weight-to-lift ratio of 1.196, indicating that the wing alone generates sufficient lift to sustain level flight. This result validates the minimum required velocity estimated in Section 1.2 and confirms that the wing design is appropriate for efficient cruise flight.

7.2.2 ASYMMETRIC PROPELLER BEHAVIOR AND TORQUE GENERATION

An important finding emerges from the examination of individual propeller thrust values: assuming equal thrust contributions from the left and right propellers at identical throttle settings is incorrect. As shown in Tables 7.1–7.3, significant differences exist between the thrust generated by the left and right propellers, even when commanded at the same throttle percentage. This asymmetry produces a non-zero torque about the drone's center of mass, which will induce a pitching moment and create flight instability if not compensated by an appropriate control strategy.

The torque values presented in the tables quantify this effect and highlight the necessity for differential throttle control to maintain stable flight during the transition phase.

7.3 CONVERGENCE ASSESSMENT

Tables 7.5–7.7 summarize the convergence behavior for each simulation case. Not all simulations achieved strict residual convergence within the prescribed tolerance. For cases that did not converge, the relevant quantities (L_2, T_{left}, T_{right}) were time-averaged over the final 500 iterations and the corresponding standard deviations are reported.

The standard deviation values provide insight into the reliability of the results. Cases marked as "CONV" or exhibiting standard deviations below 5 indicate that the solution has reached a stable quasi-steady state, and the results can be considered reliable. Moderate standard deviations in the range of 5-100 suggest that some oscillations persist, but the mean values still provide a reasonable estimate of the flow behavior. High standard deviations exceeding 100 indicate significant oscillations, suggesting that the flow has not stabilized. These results should be interpreted with caution, as they may not accurately represent the true steady-state behavior, if such a state exists for the given operating conditions.

Notable examples of high standard deviation occur at 3 m/s with 45% throttle, where the left propeller thrust exhibits a standard deviation of 616.06, and at 6 m/s with 100% throttle, where the standard deviation reaches 1056.00. These cases suggest that the interaction between the low-velocity freestream and high propeller thrust creates a highly unsteady flow field that cannot be adequately captured under the steady-state assumption.

7.4 DISCUSSION OF NON-CONVERGENCE

The difficulty in achieving convergence for certain operating conditions can be attributed to the inherently complex flow physics of the wing–propeller interaction. Flow instabilities arise naturally when a low-velocity horizontal flow is superimposed with a strong vertical momentum flux from the propellers, creating flow structures that are fundamentally unsteady, particularly at intermediate throttle settings where neither flow component dominates. The propeller wake impinges on the wing surface, creating localized regions of flow separation and reattachment that may exhibit oscillatory behavior. Furthermore, the strong coupling between the pressure jump imposed by the propellers and the local velocity field around the wing makes the iterative solution process more sensitive to residual oscillations.

While a fully transient simulation would provide more accurate results for these conditions, the computational cost would be prohibitive given the large number of operating points to be analyzed. The steady-state approach with statistical averaging therefore represents a pragmatic compromise between accuracy and feasibility for the objectives of this study.

7.5 SUMMARY OF RESULTS

TABLE 7.1
Results for U_∞=3 m/s

Throttle (%)	L_1 **(N)**	L_2 **(N)**	T_{right} **(N)**	T_{left} **(N)**	T_{total}	**Weight-lift ratio**	**Torque (Nm)**
45	1.02	1.99	3.71	0.64	7.36	0.375	0.936
55	1.02	2.94	1.28	0.95	6.19	0.315	0.101
60	1.02	3.37	1.40	1.42	7.20	0.367	-6×10^{-3}
90	1.02	5.76	3.08	4.02	13.88	0.707	-0.329
100	1.02	5.49	3.59	4.97	15.08	0.769	-0.483

TABLE 7.2
Results for U_∞=6 m/s

Throttle (%)	L_1 **(N)**	L_2 **(N)**	T_{right} **(N)**	T_{left} **(N)**	T_{total}	**Weight-lift ratio**	**Torque (Nm)**
20	4.60	1.00	0.19	0.20	5.98	0.305	3×10^{-3}
50	4.60	3.01	-0.41	0.25	7.45	0.380	-0.201
70	4.60	4.77	0.32	-0.56	9.13	0.465	0.268
80	4.60	5.86	0.62	0.06	11.13	0.567	0.171
100	4.60	6.41	5.41	1.51	17.92	0.913	1.190

7.5.1 CONVERGENCE STATISTICS

TABLE 7.3
Results for U_∞=9 m/s

Throttle (%)	L_1 **(N)**	L_2 **(N)**	T_{right} **(N)**	T_{left} **(N)**	T_{total}	**Weight-lift ratio**	**Torque (Nm)**
20	10.71	2.26	0.23	0.24	13.44	0.685	-3×10^{-3}
30	10.71	2.66	1.05	1.03	15.45	0.787	-6×10^{-3}
35	10.71	2.75	1.38	1.24	16.08	0.820	0.0427
40	10.71	2.55	1.31	1.30	15.88	0.809	3×10^{-3}
50	10.71	1.77	-0.34	1.67	13.81	0.704	-0.613
80	10.71	1.53	-1.22	1.31	12.33	0.628	-0.772

TABLE 7.4
Results for U_∞=12 m/s

Throtle (%)	L_1 **(N)**	L_2 **(N)**	T_{right} **(N)**	T_{left} **(N)**	T_{total}	**Weight-lift ratio**	**Torque (Nm)**
0	19.39	4.08	0	0	23.47	1.196	0

TABLE 7.5
Results for U_∞=3 m/s

Throttle (%)	STD L_2	STD T_{left}	STD T_{right}
45	2.88	616.06	19.25
55	4.48	165.00	39.00
60	4.60	228.40	35.68
90	CONV		
100	CONV		

TABLE 7.6
Results for U_∞=6 m/s

Throttle (%)	STD L_2	STD T_{left}	STD T_{right}
20	CONV		
50	7.54	90.75	52.74
70	9.90	218.00	99.92
80	11.18	490.00	104.57
100	10.73	1056.00	79.60

TABLE 7.7
Results for U_∞=9 m/s

Throttle (%)	STD L_2	STD T_{left}	STD T_{right}
20	CONV		
30	CONV		
35	CONV		
40	CONV		
50	1.36	103.00	7.16
80	0.72	64.00	9.46

CHAPTER 8

MESH SIZE AND DOMAIN SIZE CONVERGENCE STUDY

8.1 CRITERION

As explained previously in Section 6.3, the simulation stopping criteria are either the reduction of the residuals to an order of 10^{-6} or the attainment of the maximum number of iterations, set to 2000.

8.2 PRESENTATION OF THE DIFFERENT MESHES

Mesh	Number of elements	Number of nodes
1	6 372	6 513
2	12 936	13 136
3	25 866	26 148
4	51 546	51 945
5	102 663	103 227
6	204 395	205 192
7	410 096	411 225

TABLE 8.1
Mesh sizes used for the grid convergence study.

The specifications of each mesh are identical to those described in Section 5.3. That is, the same bias factor (which are shown id details in the second Appendix) is applied for all meshes in order to preserve similar cell growth rates and mesh topology across refinement levels.

All meshes therefore share the same overall structure and refinement strategy; only the number of divisions along each edge is increased by a factor of $1/\sqrt{2}$ between successive meshes. As a result, the characteristic cell size is reduced by a factor of $\sqrt{2}$, which is consistent with standard grid convergence practices. This refinement strategy ensures a systematic and uniform increase in spatial resolution while maintaining geometric similarity between meshes, allowing discretization errors to be isolated and the grid convergence behavior of the solution to be properly assessed.

8.2.1 MESH 1 : THE COARSEST

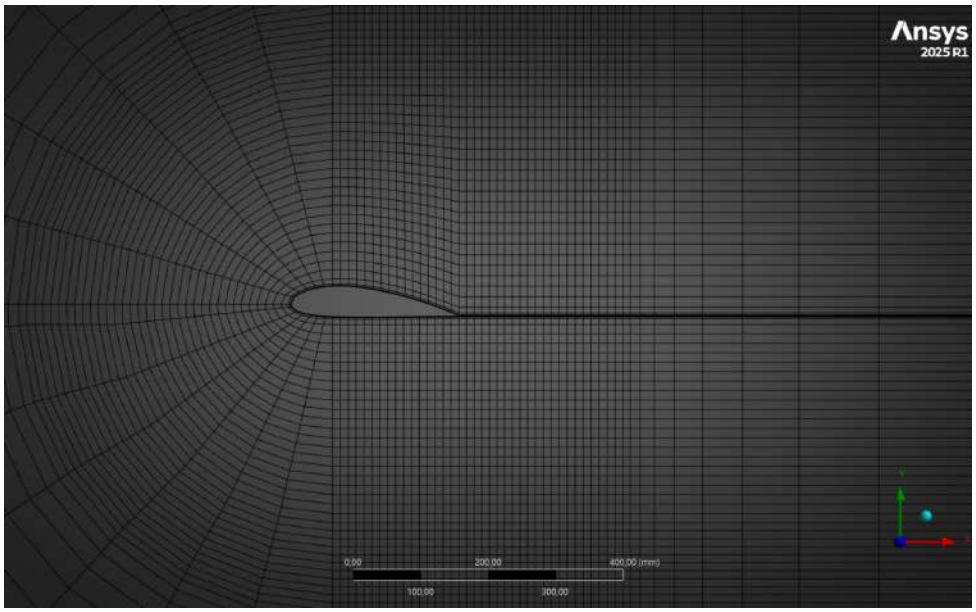


FIGURE 8.1
Mesh 1

8.2.2 MESH 2

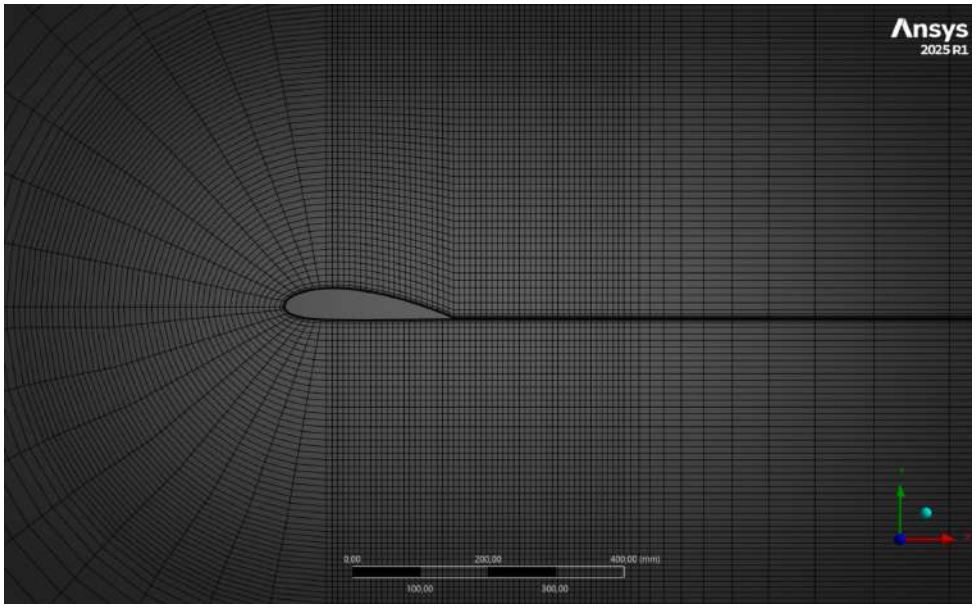


FIGURE 8.2
Mesh 2

8.2.3 MESH 3

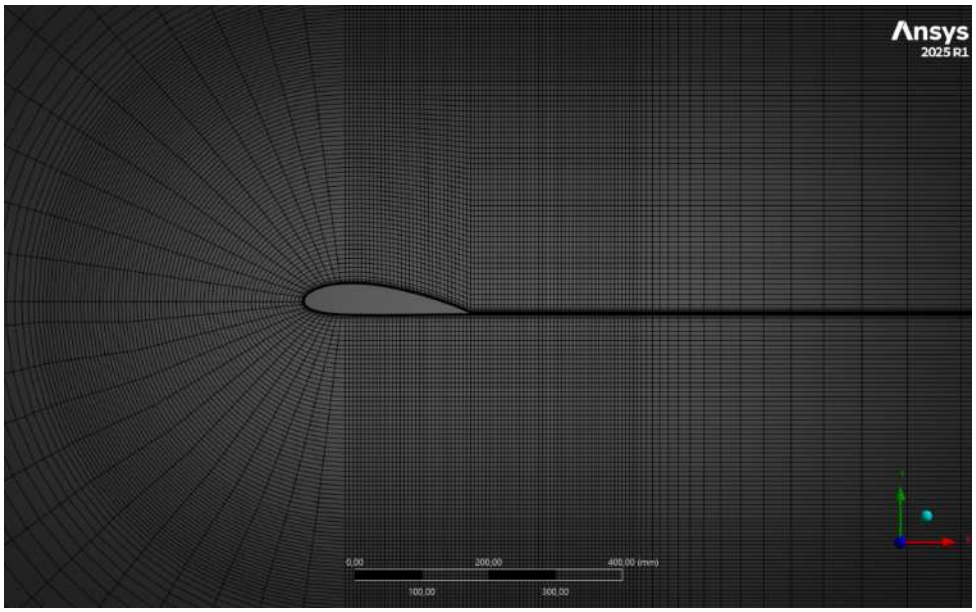


FIGURE 8.3
Mesh 3

8.2.4 MESH 4

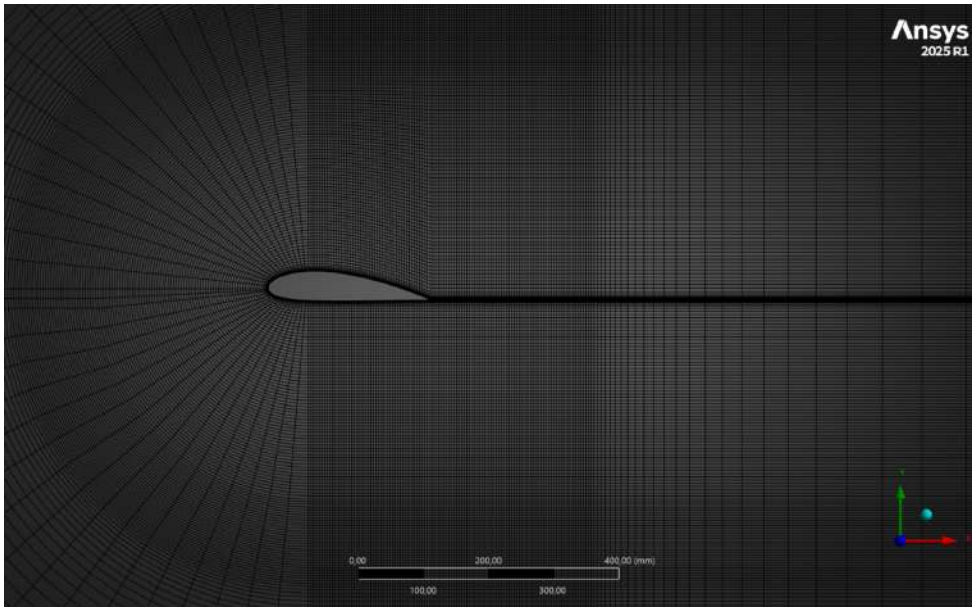


FIGURE 8.4
Mesh 4

8.2.5 MESH 5

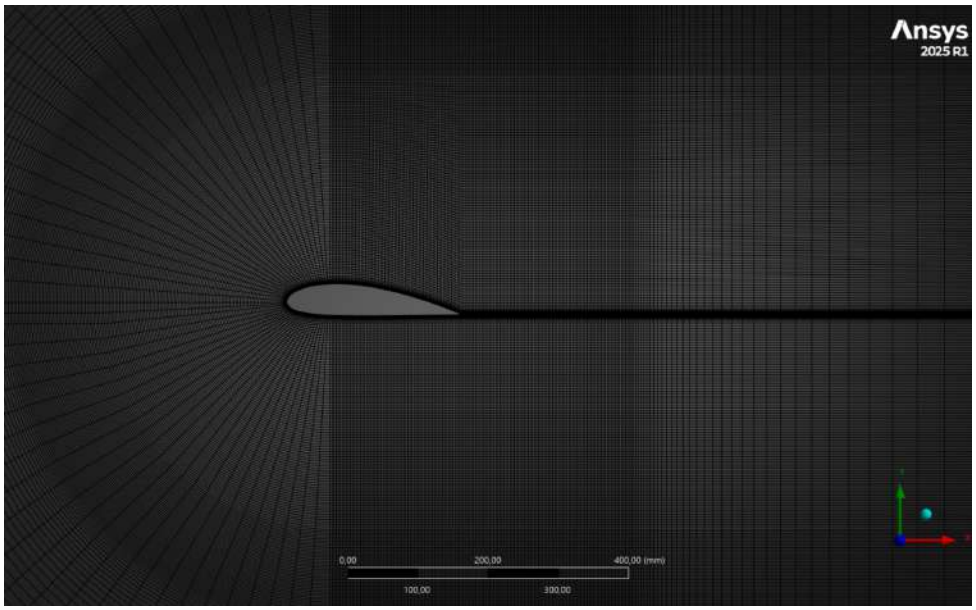


FIGURE 8.5
Mesh 5

8.2.6 MESH 6

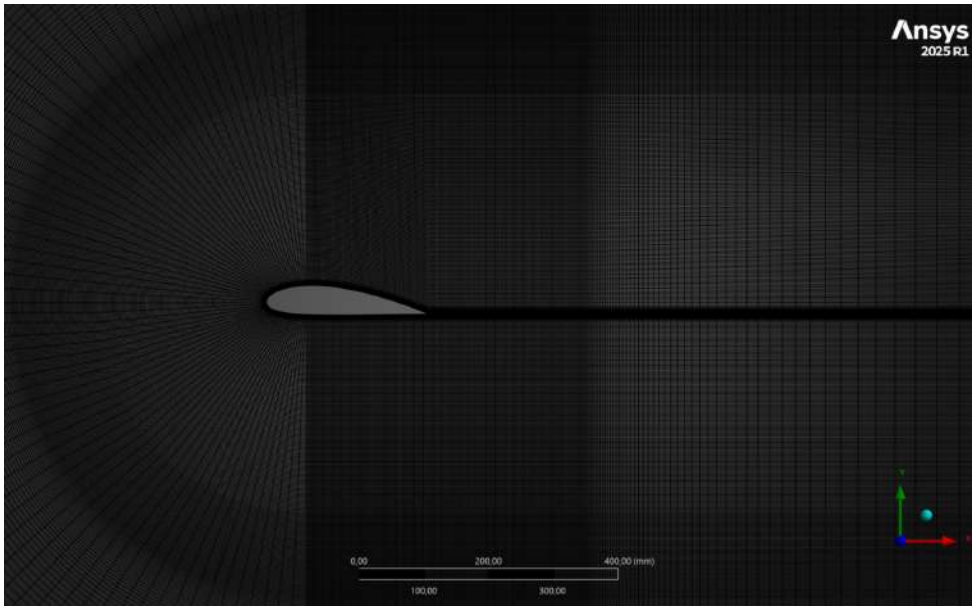


FIGURE 8.6
Mesh 6

8.2.7 MESH 7 : THE FINEST

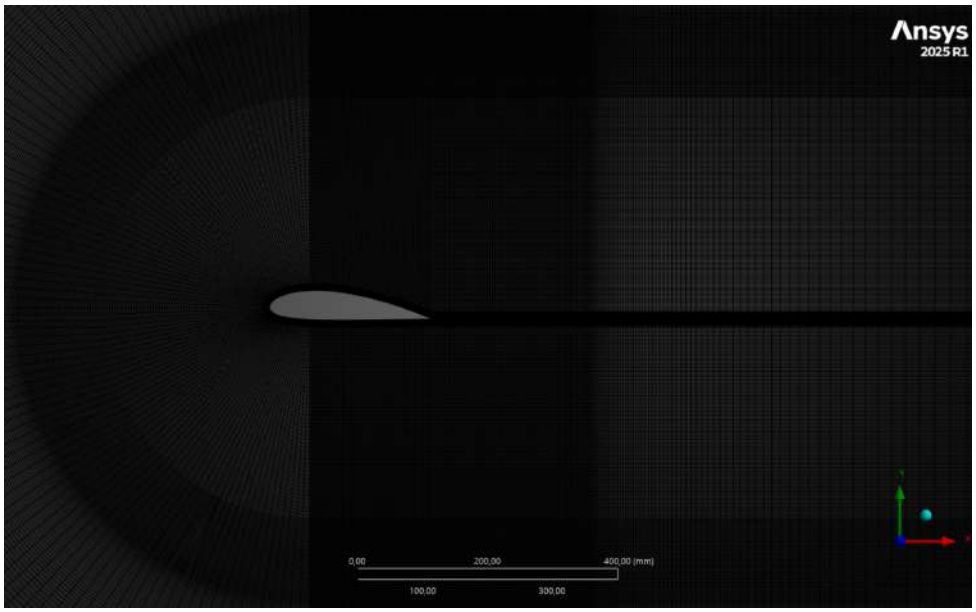


FIGURE 8.7
Mesh 7

8.3 MESH CONVERGENCE STUDY

A mesh convergence study using the seven different meshes is performed for:

1. an upstream velocity of $U_\infty = 9\ \mathrm{m/s}$ with a propeller throttle of 35%.
2. an upstream velocity of $U_\infty = 12\ \mathrm{m/s}$ with a propeller throttle of 0%.

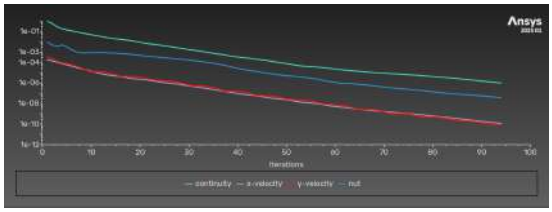
These are two cases in which the simulations converge and a steady-state solution is achieved.

8.4 RESULT ON THE COARSEST MESH

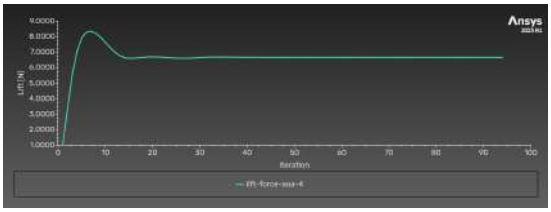
8.4.1 $U = 9\ \mathrm{m/s}$ AND 35% PROPELLER THROTTLE

As shown in Figure 8.8 (A), the residuals exhibit a satisfactory convergence behavior. It can be observed that the residual associated with the y-velocity component (y-momentum) displays a recurring concave "bump" pattern. This behavior is related to the vertical velocity component induced by the propellers in the system, which introduces oscillatory effects in the y-momentum equation during the iterative process.

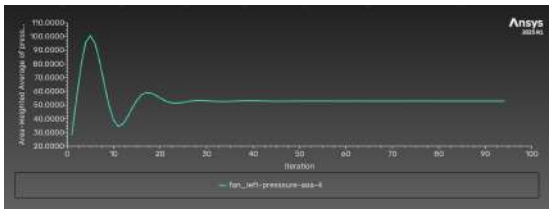
In Figure 8.8 (B), (C) and (D), the monitored physical quantities converge rapidly and reach stable values within approximately 20 to 30 iterations, indicating a well-behaved and numerically stable solution.



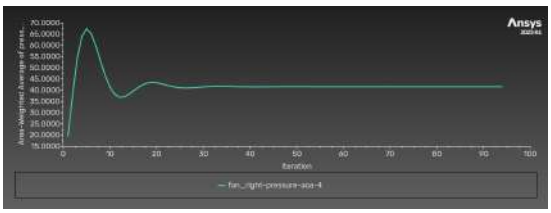
(A) Residuals convergence



(B) Lift force convergence



(C) Left fan pressure jump convergence



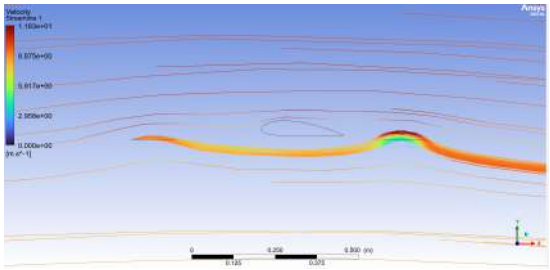
(D) Right fan pressure jump convergence

FIGURE 8.8
Results of the convergence for an upstream velocity of $U = 9\ \mathrm{m/s}$ and a propeller throttle of 35% on the coarsest mesh.

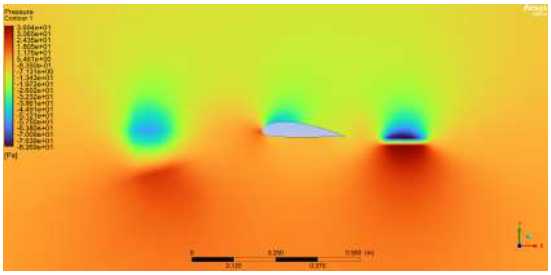
As shown in Figure 8.9 (A), the streamlines reveal a strong influence of the horizontal velocity component. In particular, the flow exiting the left propeller is rapidly redirected toward the horizontal direction and subsequently enters the right propeller. This interaction leads to a local deceleration of the flow and a corresponding increase in pressure in the region below the wing and propellers, as illustrated in Figure 8.9 (B).

In the same Figure, it is also interesting to note the shape of the pressure field beneath the left propeller, which appears tilted when compared to that beneath the right propeller. This asymmetry may be attributed to the local mesh quality, as illustrated in Figure 8.1, where the cells located below and above the left propeller are highly stretched and tilted. Such cell elongation can locally affect the accuracy of pressure gradient computations, potentially leading to distortions in the pressure field and introducing errors in the calculated quantities.

It is also interesting to note that the high-pressure region on the pressure side of the airfoil appears more pronounced than in the case of an airfoil immersed in a uniform free stream. This behavior may be attributed to the vertical velocity component injected into the flow by the left propeller, which locally slows down the airflow beneath the airfoil and consequently leads to an increase in static pressure in this region, resulting in an enhanced lift contribution. This interpretation is consistent with the simulation results, which show that the sectional lift of the wing portion affected by the propellers is higher than that of the undisturbed portion. Specifically, for an upstream velocity of $U_\infty = 9\ \mathrm{m/s}$, the lift per unit span of the undisturbed wing is $L_{1,b} = 8.64\ \mathrm{N/m}$, whereas the lift per unit span of the disturbed wing reaches $L_{2,b} = 10.58\ \mathrm{N/m}$ (value resulting from the finest mesh).



(A) Streamlines

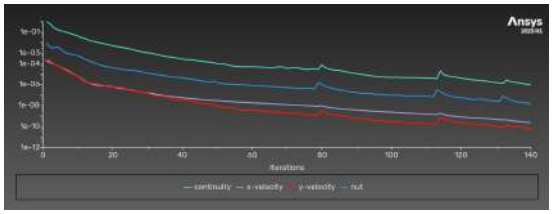


(B) Contour plot of the pressure field

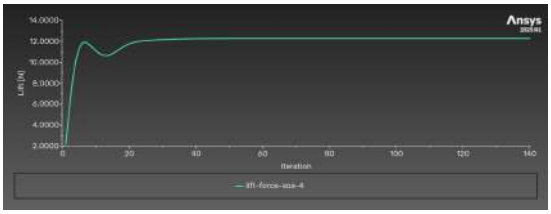
FIGURE 8.9
Results for an upstream velocity of $U = 9\ \mathrm{m/s}$ and a propeller throttle of $35\ \%$ on the coarsest mesh.

8.4.2 $U = 12\ \mathrm{m/s}$ AND 0% PROPELLER THROTTLE

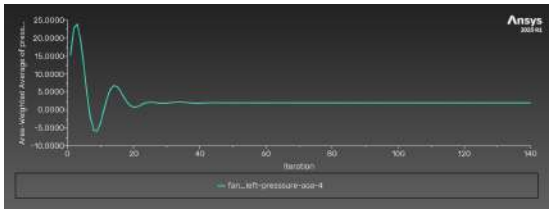
As shown in Figure 8.10 (A), the residuals exhibit a satisfactory convergence behavior. For the monitored computed quantities, shown in Figure 8.10 (B), (C) and (D), a rapid convergence is observed, similar to that reported in Section 8.4.1, with stable values reached after approximately 20 to 40 iterations.



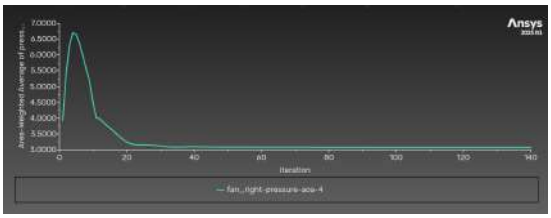
(A) Residuals convergence



(B) Lift force convergence



(C) Left fan pressure jump convergence



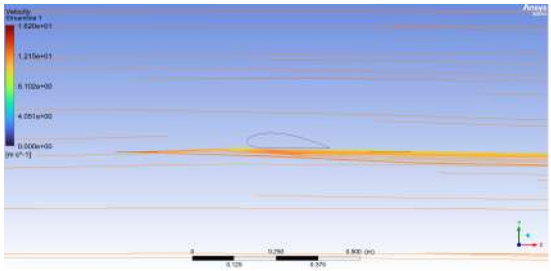
(D) Right fan pressure jump convergence

FIGURE 8.10
Results of the convergence for an upstream velocity of $U = 12\ \mathrm{m/s}$ and a propeller throttle of 0% on the coarsest mesh.

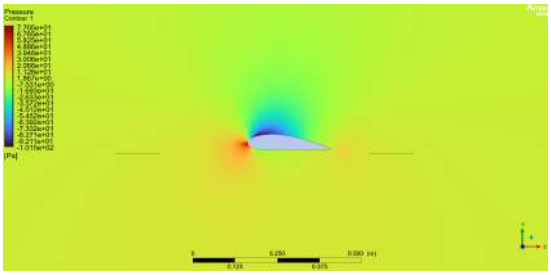
As shown by the streamlines in Figure 8.11 (A), the flow behaves as if no propeller boundary conditions were present upstream or downstream of the airfoil, which is expected since the propeller throttle is set to 0%. In Figure 8.11 (B), the pressure field distribution is characteristic of an airfoil immersed in a uniform free stream without surrounding perturbing bodies, with a clear low-pressure region on the suction side and a higher pressure on the pressure side.

However, this representation does not fully reflect the physical reality. In practice, the propellers have a finite thickness and would therefore deflect the incoming flow, potentially degrading the aerodynamic efficiency of the wing. As a consequence, a more distorted pressure field distribution would be expected

compared to the idealized pattern observed here. This limitation arises from the assumptions adopted in the modeling of the propeller boundary conditions within the present two-dimensional framework.



(A) Streamlines



(B) Contour plot of the pressure field

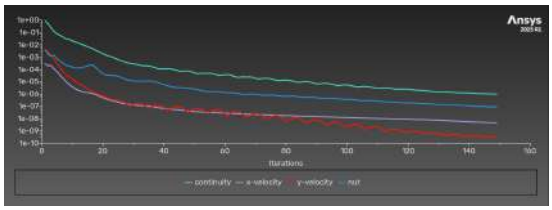
FIGURE 8.11
Results for an upstream velocity of $U = 12\ \mathrm{m/s}$ and a propeller throttle of $0\ \%$ on the coarsest mesh.

8.5 RESULT ON THE FINEST MESH

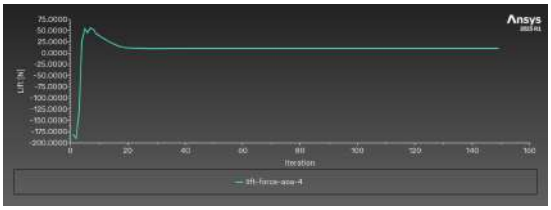
8.5.1 $U = 9\ \mathrm{m/s}$ AND 35% PROPELLER THROTTLE

With a finer mesh, the oscillatory "bump" patterns previously observed for the coarsest mesh are recovered and are now also reflected in the continuity equation residual, as shown inf Figure 8.12 (A). Such behavior is typical of strongly coupled flows, where pressure and velocity fields continuously adjust during the iterative process. On finer meshes, these interactions are resolved more accurately, which can lead to more pronounced residual oscillations without indicating numerical instability. As long as the residuals remain bounded and the monitored physical quantities converge toward stable values, the presence of such bumps should not compromise the validity of the solution.

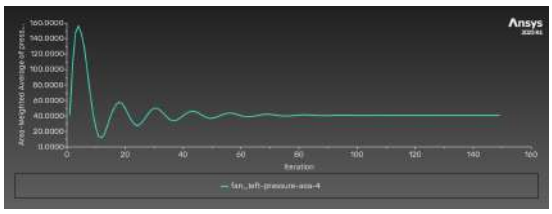
For the monitored values shown in Figures 8.12 (B), (C), and (D), the overall convergence behavior is similar to that observed with the coarser mesh. With the exception of Figure 8.12 (C), which exhibits stronger oscillations and stabilizes after approximately 80 iterations, the monitored quantities converge in a comparable manner and within a similar number of iterations.



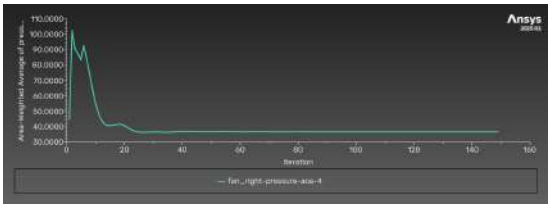
(A) Residuals convergence



(B) Lift force convergence



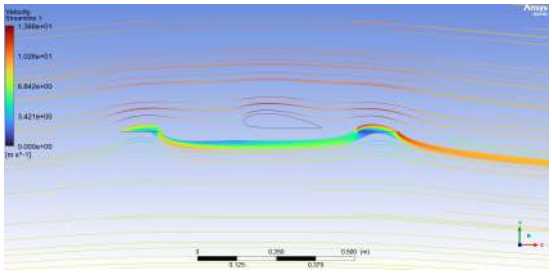
(C) Left fan pressure jump convergence



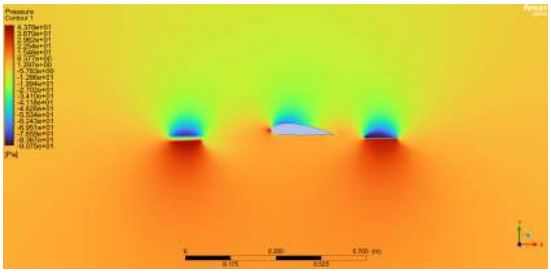
(D) Right fan pressure jump convergence

FIGURE 8.12
Results for an upstream velocity of $U = 9\ \mathrm{m/s}$ and a propeller throttle of 35% on the finest mesh.

The same observations made for the coarser mesh regarding the streamlines, in Figure 8.13 (A), also apply to the present case using the finer mesh. However, it can be noted that with the significantly refined mesh, the pressure field, in Figure 8.13 (B), beneath the left propeller no longer exhibits the previously observed tilted pattern. This indicates an improved representation of the flow field and reflects the higher accuracy of the refined mesh in resolving pressure gradients.



(A) Streamlines



(B) Contour plot of the pressure field

FIGURE 8.13
Results for an upstream velocity of $U = 9\ \mathrm{m/s}$ and a propeller throttle of $35\ \%$ on the finest mesh.

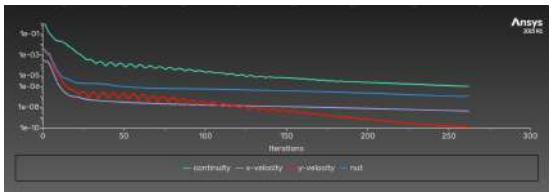
8.5.2 $U = 12\ \mathrm{m/s}$ AND 0% PROPELLER THROTTLE

For the residual convergence shown in Figure 8.14 (A), the oscillatory "bump" pattern previously observed for the case with an upstream velocity of $9\ \mathrm{m/s}$ and a propeller throttle of 35% is recovered. These bumps reflect the fact that, although the propeller boundary condition is imposed with a pressure jump of $\Delta P = 0$ for all incoming velocities, a small residual pressure jump is still computed at the boundary. This residual pressure jump injects a weak vertical velocity component into the flow, leading to small oscillations in the y-momentum equation.

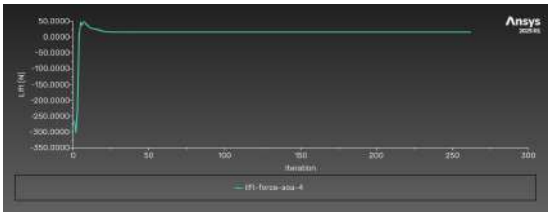
As noted during the grid convergence study (when reporting the values), even when the propeller boundary condition was set to a zero pressure jump, a residual pressure jump of the order of 2 to 3 Pa was computed

for both propellers. This contribution remains negligible from a physical standpoint, as it corresponds to a thrust force of less than 0.15 N.

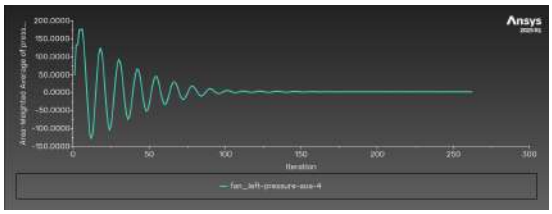
For Figures 8.14 (B), (C), and (D), the monitored quantities still exhibit satisfactory convergence behavior. Although the pressure jump associated with the left propeller, shown in Figure 8.14(C), displays slightly stronger oscillations, these oscillations progressively decay and the solution converges after approximately 150 iterations.



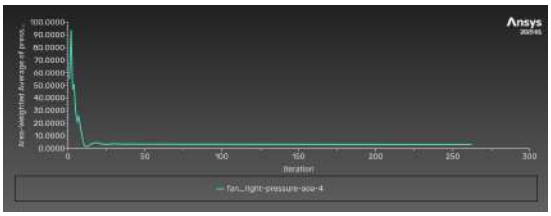
(A) Residuals convergence



(B) Lift force convergence



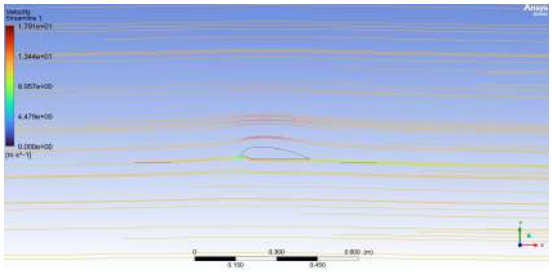
(C) Left fan pressure jump convergence



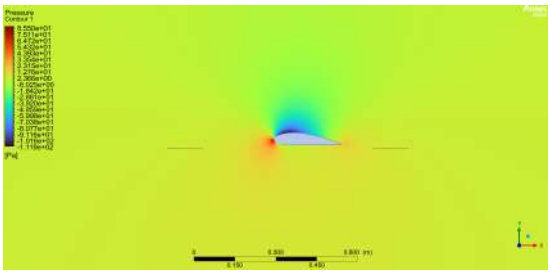
(D) Right fan pressure jump convergence

FIGURE 8.14
Results for an upstream velocity of $U = 9\ \mathrm{m/s}$ and a propeller throttle of 35%

As already discussed in Section 8.4.2, the streamlines and the pressure contour plots correspond to those expected for an airfoil immersed in a uniform free stream. In the present case, the use of a finer mesh further improves the accuracy of the flow representation. The streamlines shown in Figure 8.15 (A) and the pressure field illustrated in Figure 8.15 (B) exhibit a smoother and more physically consistent behavior, closely resembling an idealized airfoil configuration without any surrounding perturbing bodies. This behavior is consistent with the modeling assumptions adopted here, namely the absence of propeller rotation, zero imposed pressure jump and no injected vertical velocity component. Consequently, the influence of the finite thickness of the propellers is not accounted for. As a brief conclusion regarding the propeller boundary conditions, it can therefore be stated that this modeling approach does not fully represent the physical reality and remains a strong simplification with inherent limitations and another propeller model should be considered.



(A) Streamlines



(B) Contour plot of the pressure field

FIGURE 8.15
Results for an upstream velocity of $U = 12\ \mathrm{m/s}$ and a propeller throttle of $0\ \%$ on the finest mesh.

8.6 ESTIMATION OF THE RELATIVE ERROR

CASE WITH AN UPSTREAM VELOCITY OF $9\ \mathrm{m/s}$ AND A PROPELLER THROTTLE OF 35%

The grid convergence study is presented in Table 8.2. The solution obtained on the finest mesh is taken as the reference value and is assumed to provide the most accurate approximation of the true solution. For improved visualization, the relative error is plotted, in Figure 8.16, as a function of the number of elements. A clear degradation of the results can be observed starting from the third mesh and for coarser meshes, indicating that these meshes are insufficiently refined to accurately capture flow physics.

Mesh	Number of elements	Number of nodes	Total force [N]	Relative error [%]
1	6 372	6 513	15.64	2.7363
2	12 936	13 136	15.54	3.3582
3	25 866	26 148	15.60	2.9851
4	51 546	51 945	16.04	0.2488
5	102 663	103 227	16.13	0.3109
6	204 395	205 192	16.10	0.1244
7	410 096	411 225	16.08	0.0000

TABLE 8.2
Grid convergence study for $U = 9\ \mathrm{m/s}$ and a propeller throttle of 35%.

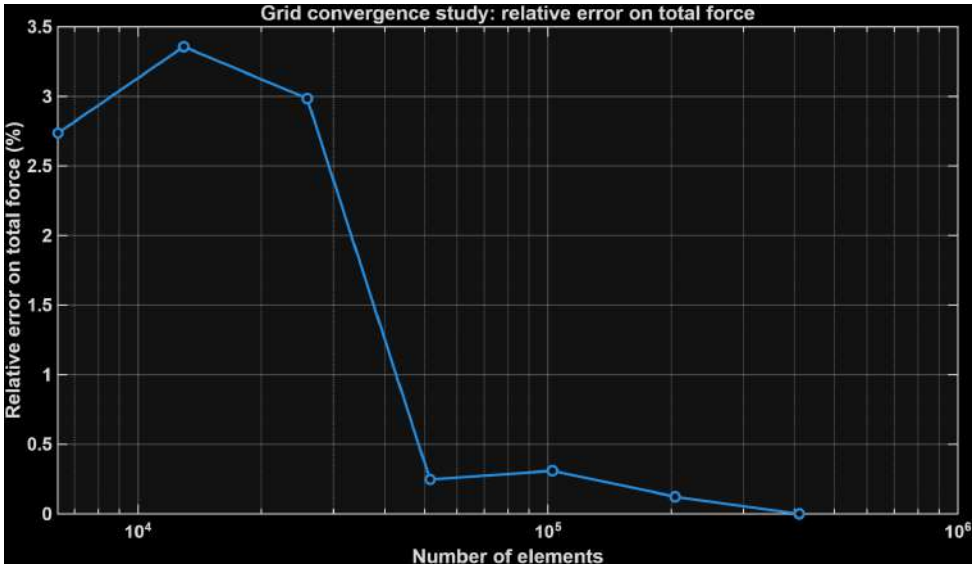


FIGURE 8.16
Convergence study performed for an upstream velocity of $U = 9\ \mathrm{m/s}$ and a propeller throttle setting of $35\,\%$.

CASE WITH AN UPSTREAM VELOCITY OF $12\ \mathrm{m/s}$ AND A PROPELLER THROTTLE OF 0%

As for the previous case, the grid convergence study is presented in Table 8.3, where the relative error is computed with respect to the solution obtained on the finest mesh, which is taken as the reference. Figure 8.17 illustrates the evolution of the relative error as a function of the number of elements. A clear degradation of the results is again observed for mesh 3 and for coarser meshes, indicating that these grids are insufficiently refined to accurately capture the relevant flow features.

Mesh	Number of elements	Number of nodes	Total force [N]	Relative error [%]
1	6 372	6 513	22.76	3.80
2	12 936	13 136	22.94	3.04
3	25 866	26 148	23.39	1.14
4	51 546	51 945	23.59	0.30
5	102 663	103 227	23.64	0.08
6	204 395	205 192	23.65	0.04
7	410 096	411 225	23.66	0.00

TABLE 8.3
Grid convergence study for $U = 12\ \mathrm{m/s}$ and a propeller throttle of 0%.

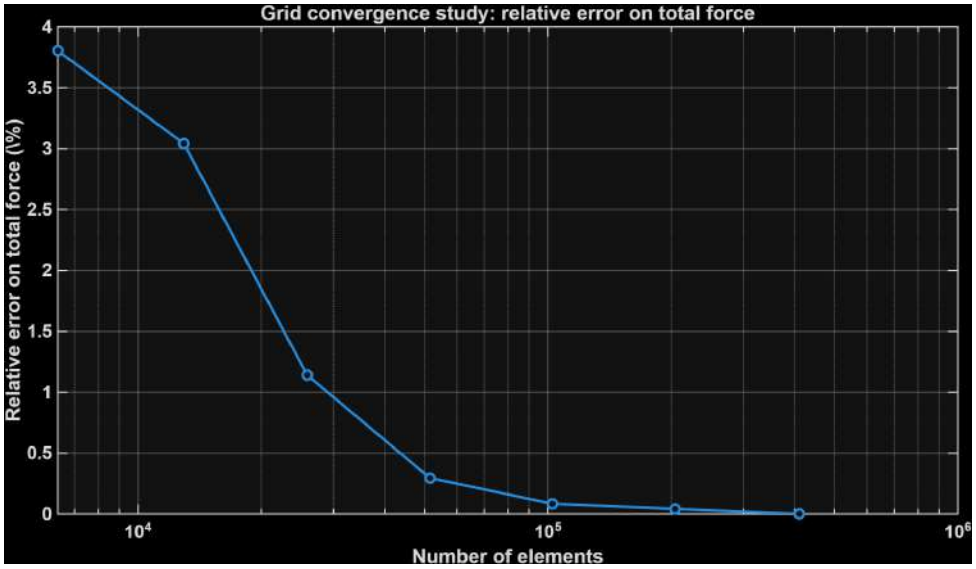


FIGURE 8.17
Convergence study performed for an upstream velocity of $U = 12\ \mathrm{m/s}$ and a propeller throttle setting of $0\,\%$.

8.7 MESH SELECTION

By examining the relative error of both cases, namely an upstream velocity of $9\ \mathrm{m/s}$ with a propeller throttle of 35%, and an upstream velocity of $12\ \mathrm{m/s}$ with zero propeller throttle, it can be observed that, for a relative error below 0.5%, mesh 4 provides the best compromise between the number of elements and the accuracy of the solution. This mesh therefore provides a grid-independent solution within the required accuracy and would have been an appropriate choice for performing the vertical-to-horizontal flight transition analysis presented in Section 7. According to standard grid convergence principles, once the discretization error falls below a prescribed tolerance, further mesh refinement yields only marginal accuracy improvements while significantly increasing the computational cost. In the present case, mesh 4 contains nearly eight times fewer elements than the finest mesh, which would have resulted in a substantial reduction of computational time with negligible loss of accuracy.

Chapter 9

Analysis and conclusions

9.1 Synthesis of Main Findings

This study investigated the aerodynamic behavior of a VTOL RC aircraft wing during the transition from vertical to horizontal flight through two-dimensional steady-state simulations across four cruise velocities (3, 6, 9 and 12 m/s). The fundamental finding is that a smooth altitude-preserving transition is not achievable with the current configuration: at low velocities (3 and 6 m/s), the combined wing lift and propeller thrust cannot support the drone's 2 kg weight even at maximum throttle (weight-to-lift ratios of 0.769 and 0.913), while at intermediate velocities (9 m/s), the throttle-to-lift relationship becomes non-monotonic due to adverse wing–propeller interaction. Only at 12 m/s does the wing alone generate sufficient lift (ratio 1.196) for sustained horizontal flight.

The grid convergence study confirmed spatial independence at mesh 4 (51,500 elements) with errors below 0.5%, though mesh 7 (410,000 elements) was used for final results. More critically, the pronounced thrust asymmetry between left and right propellers (torques exceeding 1 Nm at some conditions) reveals a fundamental control challenge: maintaining stable flight requires continuous differential throttle adjustments that would render the flow field inherently unsteady, invalidating the steady-state simulation framework itself.

9.2 The Steady-State Assumption Paradox

The most significant limitation identified is not merely the two-dimensional geometric simplification, but rather the fundamental incompatibility between the steady-state assumption and the physical reality of the transition maneuver. Active pitch control through differential throttle is essential to compensate for propeller thrust asymmetry, yet such control continuously modifies the aerodynamic boundary conditions, creating a coupled aerodynamics-flight dynamics problem where the flow field cannot reach a true steady state.

Moreover, any realistic flight control system introduces phase lags and finite response bandwidth, potentially triggering pitch oscillations precisely in the intermediate velocity regime where the non-monotonic throttle-to-lift relationship reduces control authority. The steady-state simulations cannot capture these dynamic coupling effects, suggesting that the quantitative force predictions, while internally consistent, may not represent achievable flight conditions.

9.3 CRITICAL ASSESSMENT OF PROPELLER MODELING

Despite following an established experimental approach documented in the literature for propeller boundary condition parameterization, the fan model appears to produce questionable results. The significant thrust asymmetry between geometrically identical propellers at identical throttle settings, while physically explainable through flow interaction, exhibits magnitudes that seem disproportionate. Furthermore, the non-monotonic lift behavior at intermediate velocities, where increased throttle paradoxically reduces total vertical force, suggests that the infinitesimally thin pressure-jump representation may not adequately capture the complex three-dimensional flow structures of real rotating propellers.

The absence of blade geometry, rotation-induced swirl, and finite thickness effects represents more than a quantitative simplification; it fundamentally alters the flow physics. Real propellers generate helical wake structures with strong tangential velocity components that interact differently with the wing boundary layer than the purely axial momentum injection modeled here. The experimental correlations used to derive the pressure jump may be valid for isolated propellers in uniform flow but appear to break down when applied to the highly coupled wing-propeller configuration studied here.

A more physically representative approach would require either full rotating propeller simulations using sliding mesh or actuator disk models calibrated specifically for wing-interaction scenarios, neither of which was feasible within this project's scope. The current results should therefore be interpreted as qualitatively illustrative of interaction phenomena rather than quantitatively predictive of actual flight forces.

9.4 ALTERNATIVE TRANSITION STRATEGIES

9.4.1 BALLISTIC TRANSITION WITH CONTROLLED DESCENT

Rather than attempting constant-altitude transition, a more practical approach would accept controlled altitude loss during initial acceleration. The drone would begin from sufficient altitude (15-20 m), reduce propeller thrust while pitching forward to accelerate under partial thrust and gravity, then level off once velocity reaches 9-10 m/s where wing lift becomes substantial. This converts the problem from impossible equilibrium maintenance to achievable trajectory tracking.

9.4.2 HYBRID TILT-ROTOR CONFIGURATION

Mounting vertical propellers on servo-actuated tilting mechanisms would allow gradual transition of thrust vector from vertical to horizontal as forward velocity increases, eliminating the adverse wing-propeller interaction observed in fixed configurations. While mechanically complex, this approach has proven successful in full-scale aircraft (V-22 Osprey) and would fundamentally resolve the force balance problem.

9.5 RECOMMENDATIONS FOR SIMULATION METHODOLOGY

Future work should abandon the steady-state 2D framework in favor of approaches that can capture the true coupled physics:

Three-dimensional unsteady RANS simulations with rotating propeller models (sliding mesh or MRF) are essential to capture the helical wake structure, induced drag, tip vortices, and swirl-boundary layer interaction that dominate the real flow physics. While computationally expensive (days to weeks per case), modern mesh adaptation and high-performance computing make this feasible.

Coupled CFD-flight dynamics co-simulation integrating the flow solver with six-degree-of-freedom rigid

body dynamics and closed-loop control system models would reveal whether the control-aerodynamics coupling actually leads to dynamic instability. This represents the only simulation approach capable of assessing true transition feasibility.

Reduced-order models calibrated against wind tunnel measurements offer a pragmatic alternative, sacrificing CFD accuracy for computational speed to enable extensive parametric studies and hardware-in-the-loop testing. The experimental validation would compensate for model simplifications while providing real-world confidence.

Most critically, **experimental validation through wind tunnel testing** is mandatory before any flight attempts. The propeller boundary condition modeling uncertainties identified here are too severe to permit flight clearance based on simulations alone.

9.6 KEY TAKEAWAYS DESPITE LIMITED QUANTITATIVE UTILITY

While the numerical force predictions themselves may not be reliable enough for flight control design, this study has generated several valuable insights that justify the effort:

Identification of the control-aerodynamics coupling problem: The discovery that differential throttle control fundamentally contradicts the steady-state assumption is a conceptual contribution that would not have been apparent without attempting the simulations. This finding redirects future work toward coupled simulation frameworks.

Demonstration of simulation methodology limitations: By pushing the steady-state 2D approach to its breaking point, the study clearly delineates what can and cannot be learned from simplified simulations, providing valuable guidance for resource allocation in future projects.

Qualitative characterization of flow regimes: Even if quantitatively uncertain, the simulations successfully identified three distinct aerodynamic regimes (thrust-dominated low speed, adverse interaction at intermediate speed, lift-dominated cruise) that must be addressed in any transition strategy.

Grid convergence verification protocol: The systematic mesh refinement study established a rigorous methodology applicable to future simulations and confirmed that observed phenomena are not numerical artifacts.

Physical understanding of wing–propeller interaction: The visualization of asymmetric pressure fields, flow deflection patterns and wake impingement provided physical insight into the mechanisms driving the force imbalances, informing design modifications.

9.7 CONCLUDING REMARKS

This study demonstrates that the transition from vertical to horizontal flight cannot be achieved smoothly at constant altitude with the investigated VTOL configuration, primarily due to insufficient vertical force at low velocities and destabilizing propeller thrust asymmetry requiring active control that contradicts the steady-state assumption. The propeller boundary condition modeling, while based on established approaches, appears inadequate for capturing the complex wing–propeller interaction physics.

Alternative strategies including ballistic descent, tilt-rotor configurations, or pre-programmed trajectories offer more promising paths forward than attempting to refine the current approach. Future simulation work must embrace three-dimensional unsteady methods with coupled flight dynamics or experimental validation to achieve predictive accuracy.

Despite limited quantitative reliability, the study successfully identified fundamental physical limitations and methodology boundaries that provide valuable direction for future work. The primary recommendation is clear: experimental validation through wind tunnel or flight testing is essential before any operational deployment, as simulation-based flight clearance is not justified given the modeling uncertainties identified here.

CHAPTER 10

APPENDIX

1. Experimental data of the propeller

T5147 Tri-Blade	50%	594.85	23.39	7.29	17739	170.61	3.49	82 (Ambient Temperature:25°C)
	55%	701.01	23.35	9.19	19071	214.50	3.27	
	60%	802.49	23.30	11.15	20330	259.91	3.09	
	65%	914.58	23.25	13.48	21795	313.47	2.92	
	70%	1023.23	23.19	15.91	22965	368.97	2.77	
	75%	1114.96	23.14	18.54	24177	429.06	2.60	
	80%	1208.87	23.08	21.04	24991	485.49	2.49	
	85%	1307.25	23.01	23.78	25939	547.13	2.39	
	90%	1424.98	22.94	26.84	26859	615.75	2.31	
	95%	1515.67	22.86	29.85	27829	682.49	2.22	
	100%	1730.69	22.69	36.11	29197	819.57	2.11	

FIGURE 10.1

FIGURE 10.2
tri-blade T5147 propeller

2. All the edge divisions used to create the finest mesh in section 5.4 are listed below.

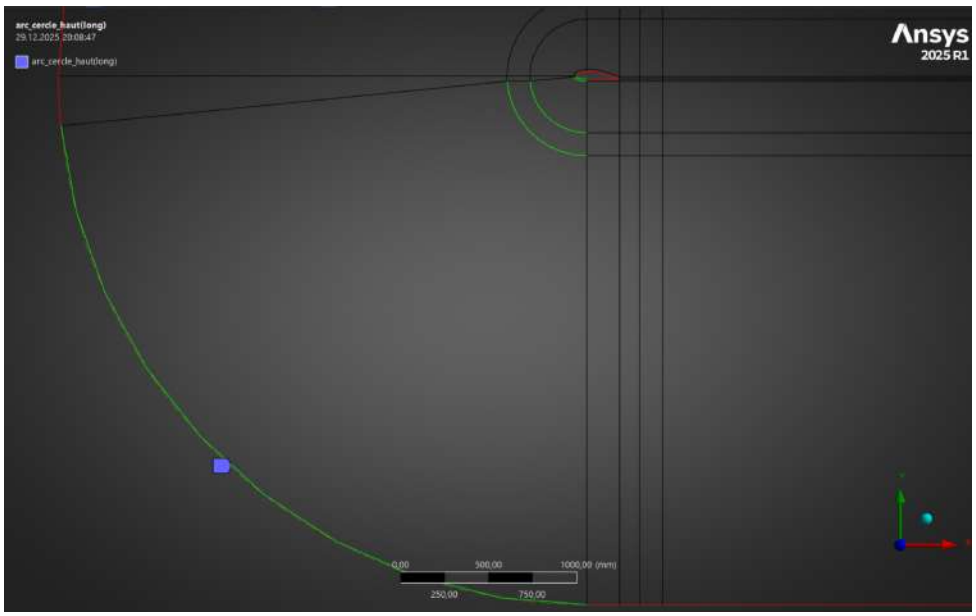


FIGURE 10.3
Number of divisions : 47

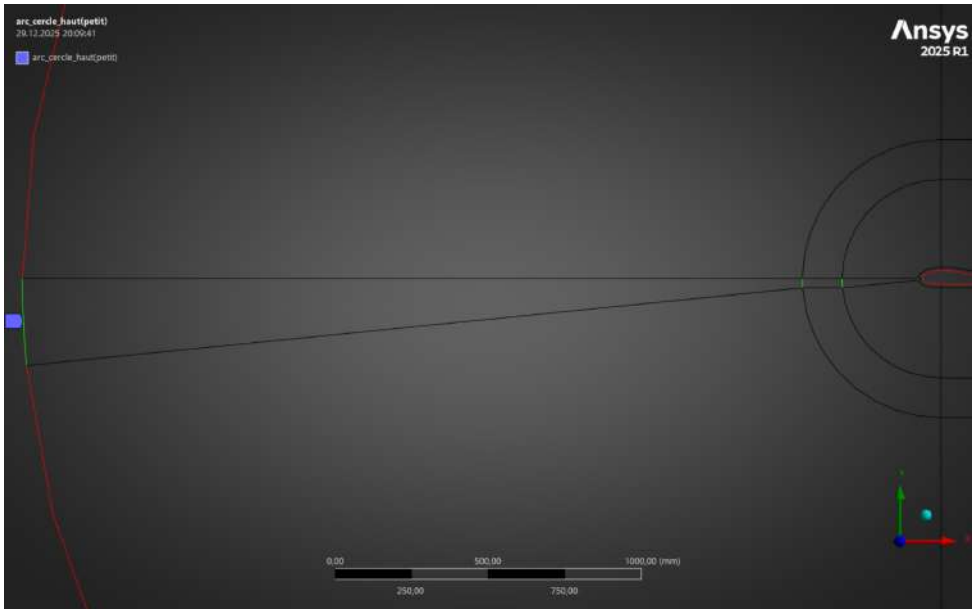


FIGURE 10.4
Number of divisions : 3

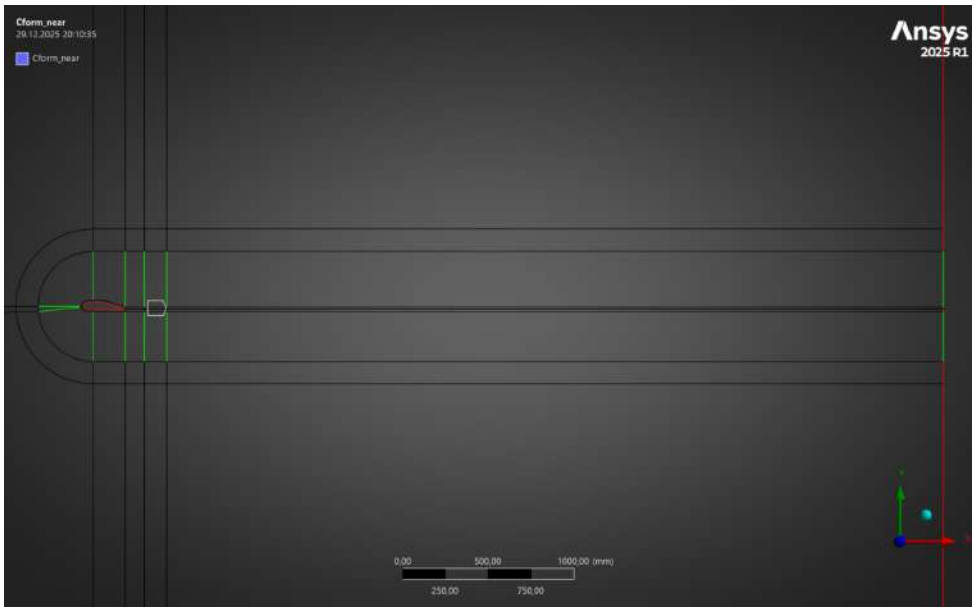


FIGURE 10.5
Number of divisions : 170

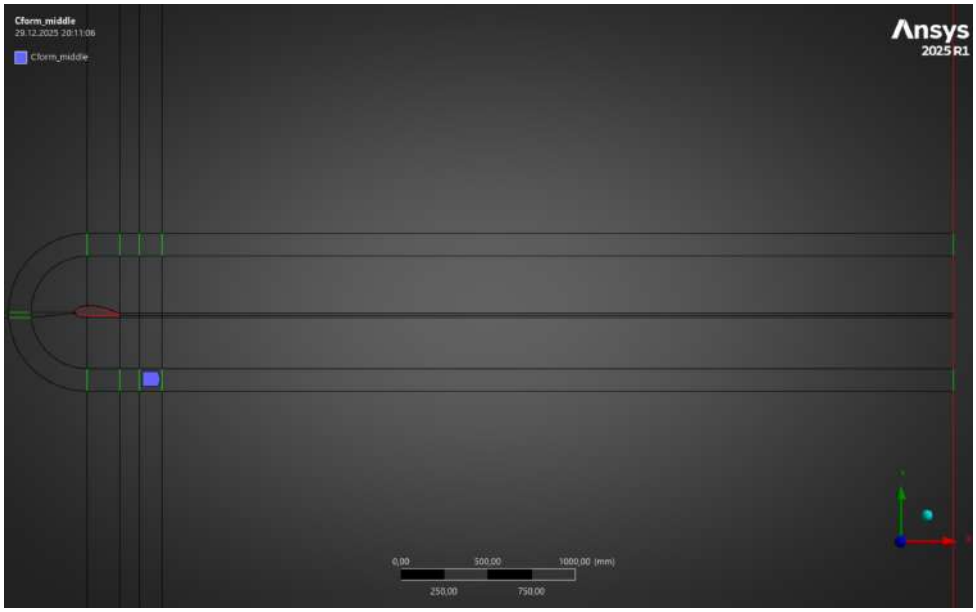


FIGURE 10.6
Number of divisions : 89

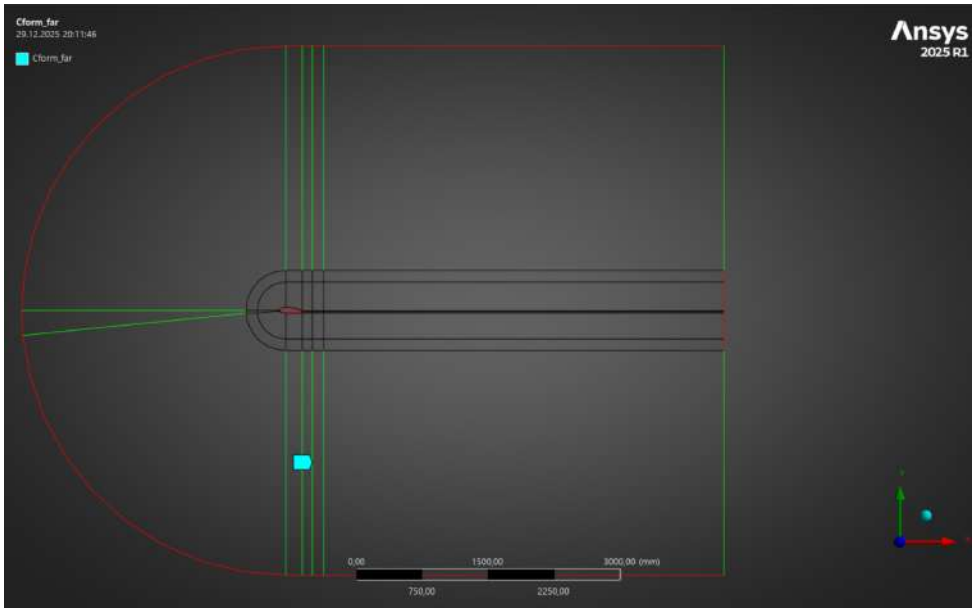


FIGURE 10.7
Number of divisions : 115 and bias factor : 60

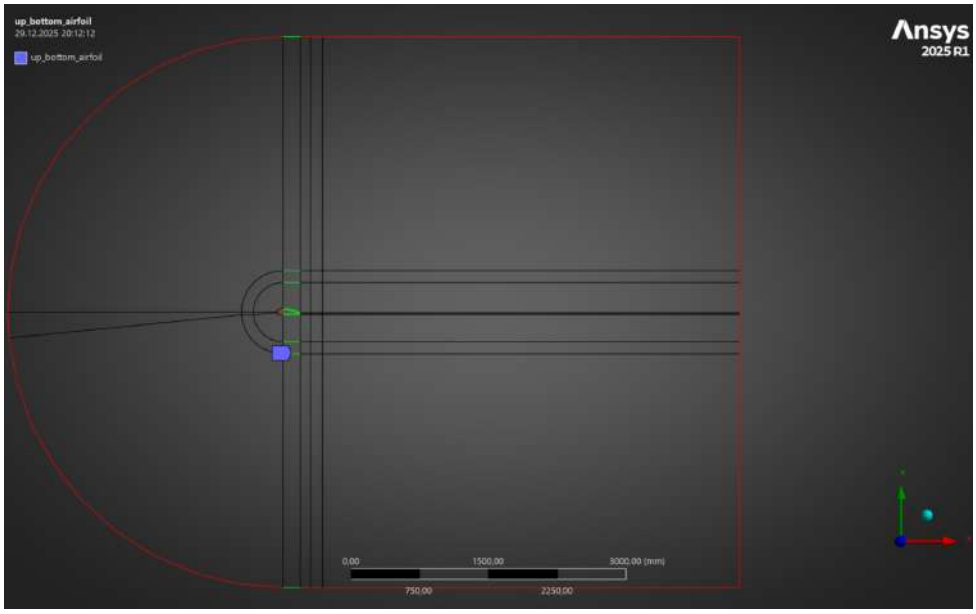


FIGURE 10.8
Number of divisions : 129

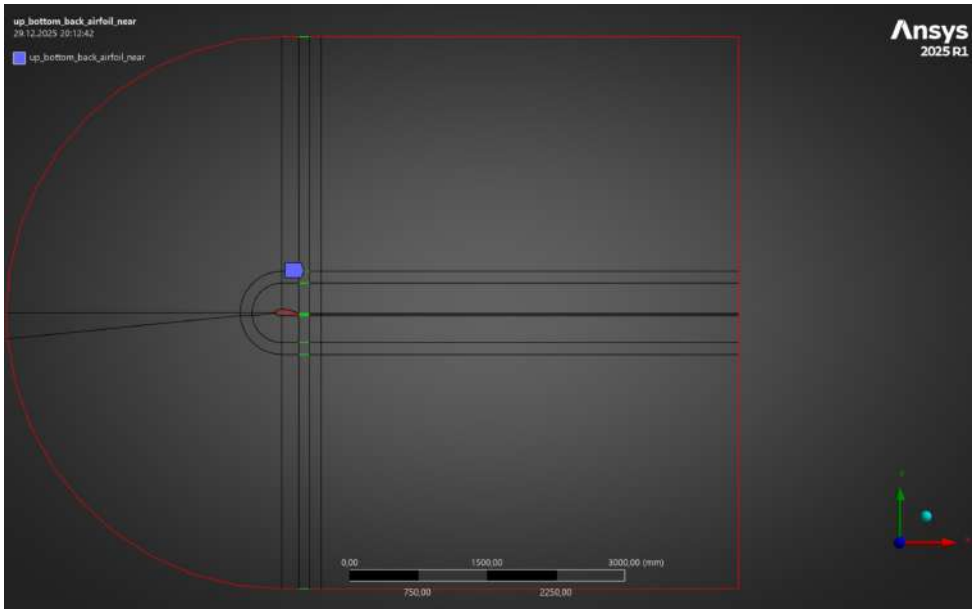


FIGURE 10.9
Number of divisions : 78

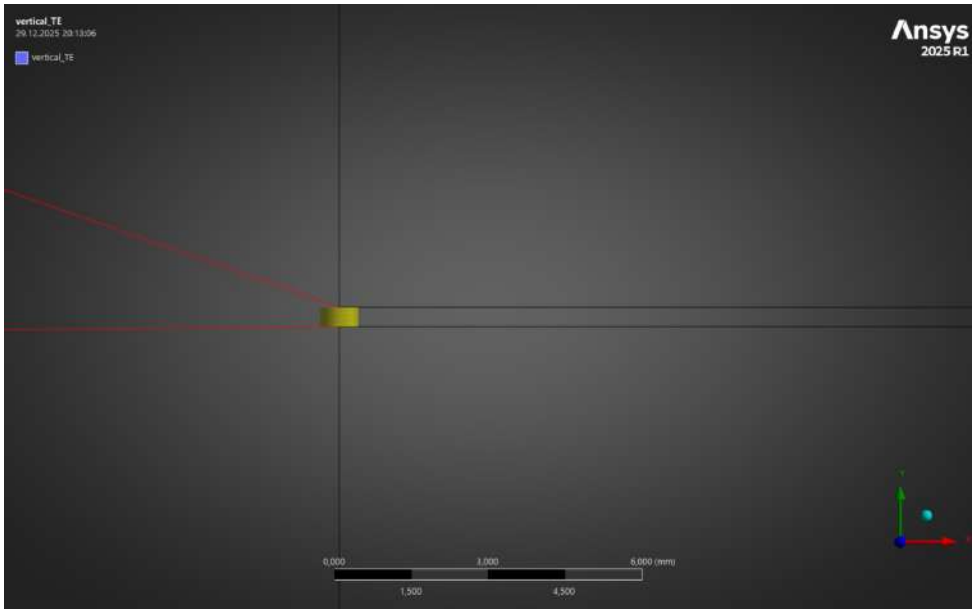


FIGURE 10.10
Number of divisions : 66

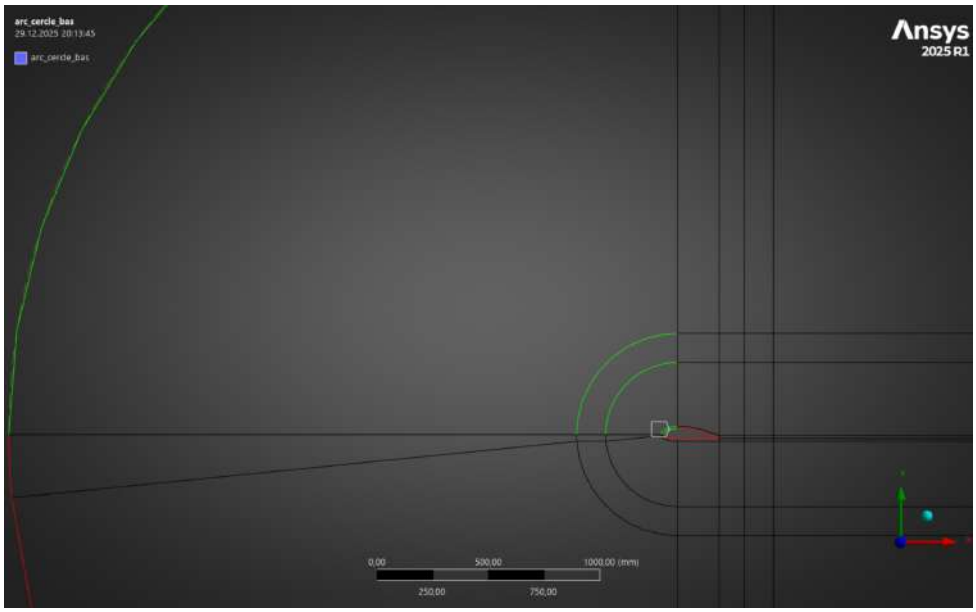


FIGURE 10.11
Number of divisions : 47

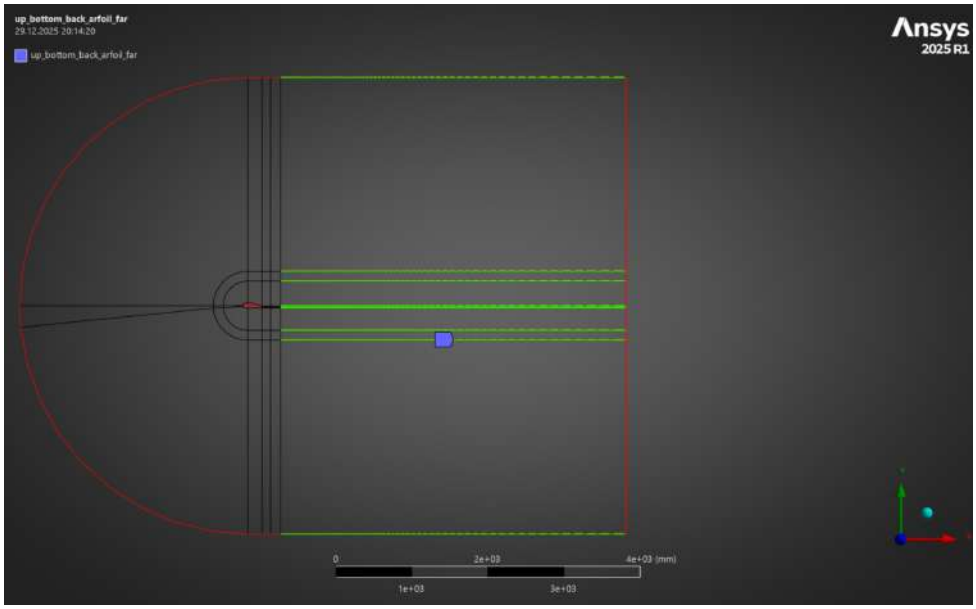


FIGURE 10.12
Number of divisions : 117 and bias factor : 128

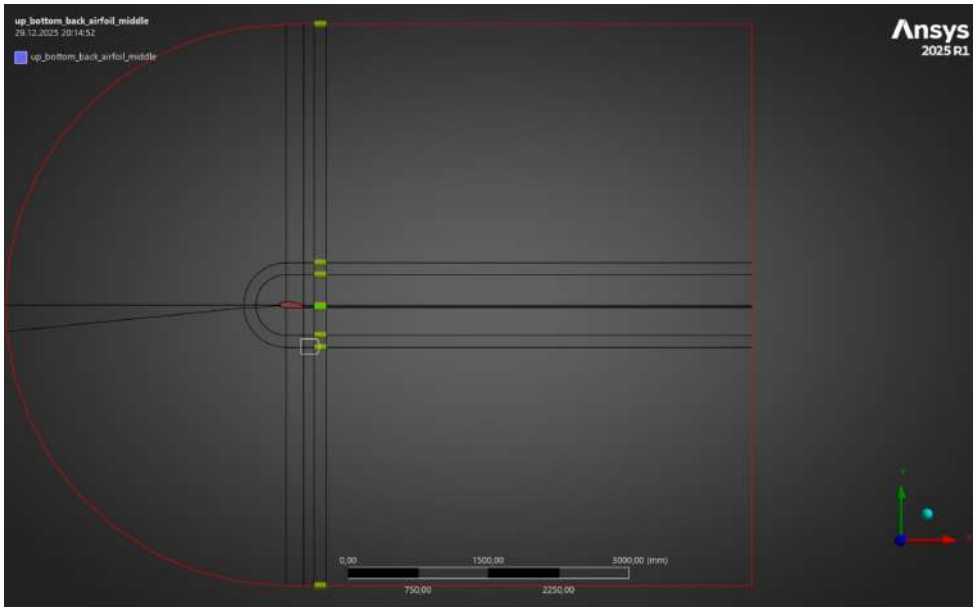


FIGURE 10.13
Number of divisions : 89

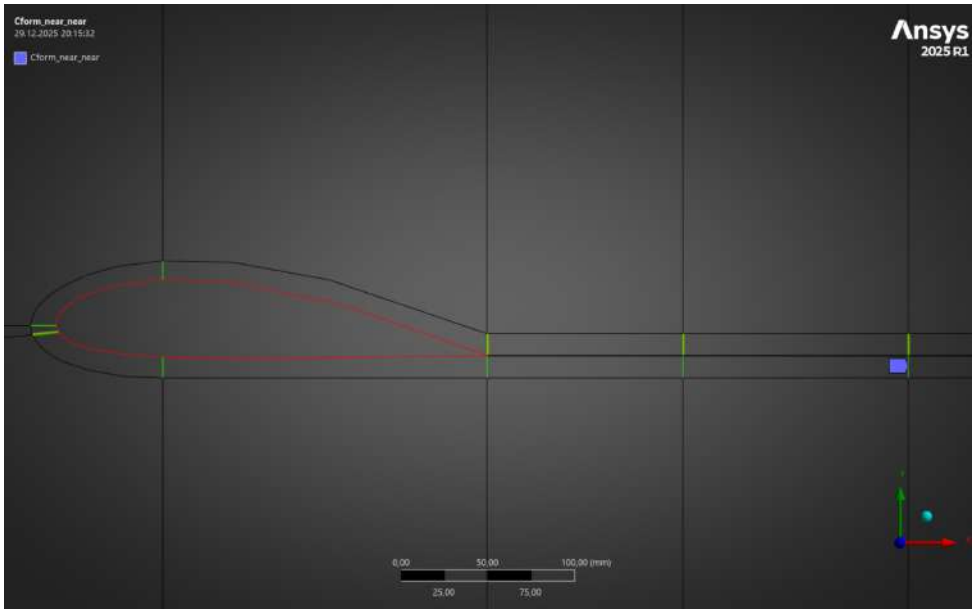


FIGURE 10.14
Number of divisions : 50 and bias factor : 250

3. Table with the design points of all the 7 meshes. The parameters from 1 to 12 correspond exactly, and in the same order, to the edge divisions presented in the second appendix.

Parameter	Mesh 1	Mesh 2	Mesh 3	Mesh 4	Mesh 5	Mesh 6	Mesh 7
1	6	8	11	16	23	33	47
2	1	1	1	2	2	2	3
3	21	30	42	60	85	120	170
4	11	16	23	32	45	63	89
5	14	20	28	40	57	81	115
6	16	23	32	45	64	91	129
7	10	14	20	28	39	55	78
8	8	11	16	23	33	47	66
9	6	8	11	16	23	33	47
10	15	21	30	42	59	83	117
11	11	16	23	32	45	63	89
12	6	9	13	18	25	35	50

TABLE 10.1
Edge division parameters for each mesh

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